Nuclear Scattering Numerical Details

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Chapter 1

Nuclear Stopping

1.1 Binary Collision Physics

In the binary collision approximation (BCA), the interaction of an energetic atom ("ion" or "projectile", index 1) with a target atom ("recoil", index 2) is treated as a sequence of isolated events, the binary collisions. In each collision, the trajectories are approximated by their asymptotes. Incoming and outgoing asymptote are the straight lines that are fitted to the trajectories long enough before and after the collision, respectively, assuming that there is no interaction with any other target atom. They are indicated in Fig. 1.1 by dashed lines. The outgoing trajectories are characterized by the scattering angle of the ion, ψ_1, the recoil angle ψ_2, and the intersection points of the asymptotes, which are given by (x_1, p) and $(x_2, 0)$ in the coordinate system of Fig. 1.1. These characteristics depend on the ion energy E before the collision, the impact parameter p (the distance of the target atom before the collision from the incoming ion asymptote), and the interatomic potential $V(r)$ which is assumed to depend on the internuclear distance only. The interatomic potential depends on the properties of the projectile and the recoil, i.e., the masses M_1 and M_2 and atomic numbers Z_1 and Z_2. During the collision, the ion transfers energy T_n to the recoil, so the ion energy after the collision is $E - T_\mathrm{n}$, and the recoil energy is T_n, if we assume the collision to be elastic and the recoil is at rest before the collision.

In this section we present the results of the classical theory of scattering for the scattering angles ψ_1 and ψ_2, for the energy transfer T_n, and for the coordinates x_1 and x_2 of the intersection points of the trajectory asymptotes, compare Fig. 1.1. Derivations can be found in Ref. [1].

In the classical theory of scattering [2, 3] the scattering angles ψ_1 and ψ_2 and the energy transfer T_n are expressed in terms of the scattering angle θ in the center-of-mass (CM) system

$$\tan\psi_1 = \frac{\sin\theta}{\cos\theta + \frac{M_1}{M_2}}, \quad \psi_2 = \frac{\pi - \theta}{2} \ , \tag{1.1}$$

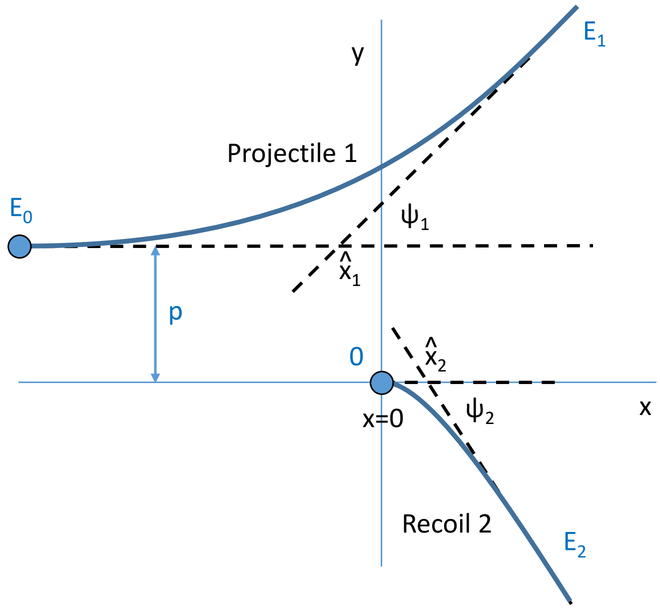

Figure 1.1: Ion and recoil trajectory and their asymptotes. The initial ion energy is E; T_n is transfered from the ion to the recoil during the collision. The energy transfer T_n and the scattering angles ψ_1 and ψ_2 depend on the initial ion energy E and the impact parameter p. The origin of the coordinate system is chosen in the intial recoil position, and the x axis is parallel to the direction of the incoming ion.

$$T_\mathrm{n} = \frac{4M_1M_2}{(M_1+M_2)^2}\,E\,\sin^2\frac{\theta}{2}\;. \tag{1.2}$$

θ is given by [4][1]

$$\pi-\theta = 2\,p\int_{r_0}^{\infty}\frac{\mathrm{d}r}{r^2\sqrt{1-\frac{V(r)}{E_\mathrm{c}}-\frac{p^2}{r^2}}}\;. \tag{1.3}$$

r_0 denotes the distance of closest approach between the ion and the recoil, which is given by the root of the expression under the square root, i.e.,

$$1-\frac{V(r_0)}{E_\mathrm{c}}-\frac{p^2}{r_0^2}=0 \tag{1.4}$$

The numerical solution of Eq. (1.4) is discussed in Section 1.3. E_c denotes the energy in the center-of-mass system

$$E_\mathrm{c} = \frac{M_2}{M_1+M_2}\,E. \tag{1.5}$$

The coordinates of the intersection points of the trajectory asymptotes may be expressed in terms of θ and the so-called time integral τ. In the coordinate system of Fig. 1.1 these

[1] We do not write θ explicitly because for very small θ this would result in a loss of accuracy in cases where actually $\pi-\theta$ is needed, if θ were first calculated by the difference of π and the integral, and then $\pi-\theta$ were calculated as the difference between π and θ. Note that $\sin\frac{\theta}{2}=\cos\frac{\pi-\theta}{2}$, $\cos\frac{\theta}{2}=\sin\frac{\pi-\theta}{2}$.

points are at $(\hat{x}_1, p)$ and $(\hat{x}_2, 0)$ with[2]

$$\hat{x}_2 = \frac{2M_1}{M_1 + M_2} \left(p \tan\frac{\theta}{2} - \tau\right) \tag{1.6}$$

and

$$\hat{x}_1 = \hat{x}_2 - p \tan\frac{\theta}{2}. \tag{1.7}$$

The time integral τ is given by [5][3]

$$\tau = r_0 - \int_{r_0}^{\infty} \left[\frac{1}{\sqrt{1 - \frac{V(r)}{E_c} - \frac{p^2}{r^2}}} - 1 \right] \mathrm{d}r. \tag{1.8}$$

Note, in spite of its name, the time integral τ has units of a length.

1.2 Scattering Quantities for Screened Coulomb Potentials

1.2.1 Screened Coulomb Potentials

For small internuclear distances r, the interaction between two atoms is described by the Coulomb potential. It is therefore common to write the interatomic potential as a screened Coulomb potential

$$V(r) = \frac{Z_1 Z_2 e^2}{r} \Phi\left(\frac{r}{a_\mathrm{I}}\right) \tag{1.9}$$

where the screening function Φ describes the deviation from the Coulomb potential and $\Phi(r \to 0) = 1$. The Coulomb potential is given by

$$V_\mathrm{Coul} = \frac{1}{4\pi\varepsilon_0} \frac{Z_1 Z_2 e^2}{r} \approx 14.3996439\,\mathrm{eV}\,\frac{Z_1 Z_2}{r[\text{Å}]} \approx 14.4\,\frac{Z_1 Z_2}{r[\text{Å}]}\,\mathrm{eV}\ . \tag{1.10}$$

The argument of Φ is usually scaled by a screening length a_I, which is chosen as to make $\Phi(r/a_\mathrm{I})$ approximately independent of the atom species. The most common choices of a_I are according to Firsov [6]

$$a_\mathrm{F} = \frac{0.46850\text{Å}}{\left(Z_1^{1/2} + Z_2^{1/2}\right)^{2/3}}\ , \tag{1.11}$$

[2] More sophisticated formulae that take electronic energy loss in the middle of the collision into account are presented in Ref. [1]. They may be used in IMSIL if desired, but the added accuracy is deemed marginal.

[3] A different formula is used in Ref. [1]. We prefer Eq. (1.8), since it is somewhat simpler to evaluate, and it can also be used for attractive interatomic potentials.

Table 1.1: Coefficients of the sum or exponentials, Eq. (1.14), for the Moliere [10], KrC [9], and ZBL potential [3].

Moliere		KrC		ZBL	
a_i	b_i	a_i	b_i	a_i	b_i
0.1	6	0.335381	1.919249	0.18175	3.1998
0.55	1.2	0.473674	0.637174	0.50986	0.94229
0.35	0.3	0.190945	0.278544	0.28022	0.4029
				0.02817	0.20162

Lindhard [7]

$$a_{\mathrm{L}} = \frac{0.46850\text{Å}}{\left(Z_1^{2/3} + Z_2^{2/3}\right)^{1/2}} , \tag{1.12}$$

and ZBL [3]

$$a_{\mathrm{ZBL}} = \frac{0.46850\text{Å}}{Z_1^{0.23} + Z_2^{0.23}} . \tag{1.13}$$

For the screening function Φ, a sum of exponentials

$$\Phi(R) = \sum_{i=1}^{n} a_i \exp(-b_i R) \tag{1.14}$$

is often used, where a_i and b_i are coefficients. The requirement $\Phi(0) = 1$ results in $\sum_i a_i = 1$. The values of the coefficients for common potentials potential are listed in Table 1.1. The Moliere [8] and KrC [9] screening function are meant to be used with the Firsov screening length, while the ZBL scrrening function [3] is used with the ZBL screening length.

Atom-pair specific interatomic potentials have also been proposed: The coefficients for pair-specific ZBL interatomic potentials are contained in the POTCOEF03.DAT file of the Data folder of SRIM-2013 [11].[4] More recently, pair-specific NLH potentials [12] have been proposed, which are fits to all-electron density functional theory (DFT) calculations.[5]

The sum-of-exponentials screening function, Eq. (1.14), has infinite range. In reality, the interatomic potential goes to zero at some internuclear distance r_{max}, and is negative for $r > r_{\mathrm{max}}$, resulting in an attractive part of the potential. The DFT data can be well fitted by a combination of a sum-of-exponentials and a linear function up to $r = r_{\mathrm{max}}$ [13]

$$\Phi(R) = \sum_{i=1}^{n} a_i \left[\exp(-b_i R) - \exp(-b_i R_{\mathrm{max}})\right] - d \left(1 - \frac{R}{R_{\mathrm{max}}}\right) , \tag{1.15}$$

[4] According to the version history that comes up when pushing the "SRIM-2013.00" button in the SRIM-2013 Main Window, the pair-specific potentials seem to have been introduced in SRIM-2003.00. They are most likely not used by the Monte Carlo TRIM code of SRIM.

[5] The NLH potentials are not meant to be used for binary collision approximation simulations as they have not been fitted to the low-energy parts of the interatomic potential.

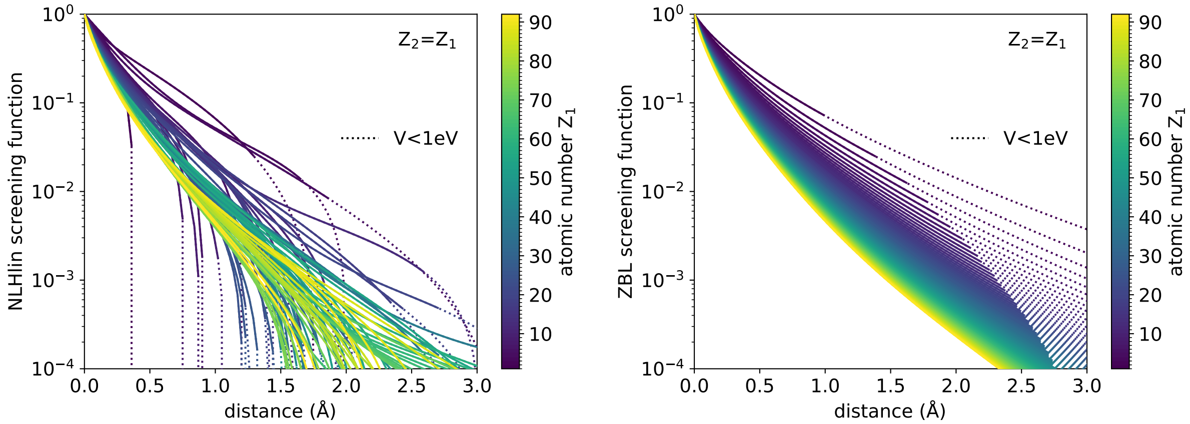

Figure 1.2: NLHlin (left) and ZBL (right) screening function for homonuclear atom pairs ($Z1 = Z_2$) as a function of internuclear distance in Ångstøms.

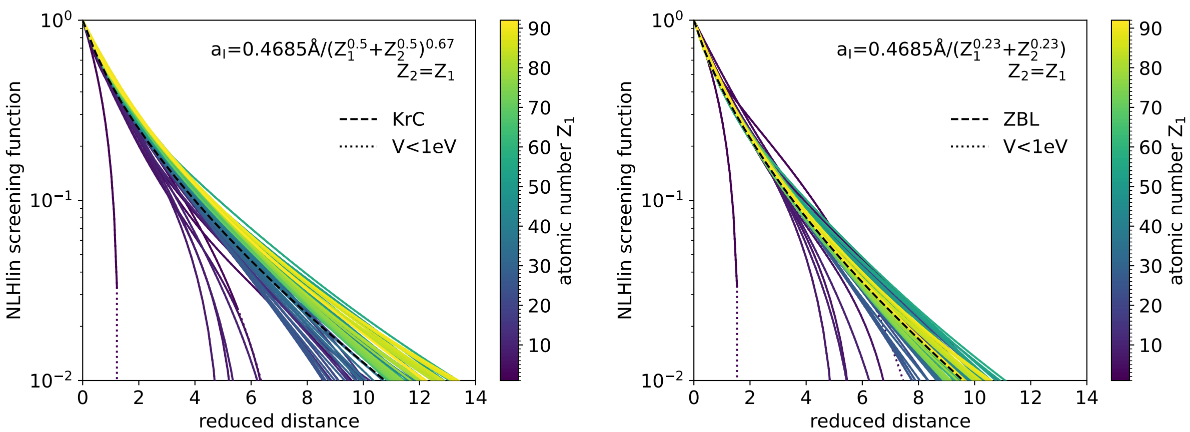

Figure 1.3: NLHlin screening function for homonuclear atom pairs ($Z_1 = Z_2$) as a function of internuclear distance scaled with the Firsov screening length Eq. (1.11) (left) and the ZBL screening length Eq. (1.13) (right).

where $R_{\mathrm{max}} = r_{\mathrm{max}}/a_{\mathrm{I}}$. A simple choice for $r > r_{\mathrm{max}}$ is $\Phi(R) = 0$, resulting in a finite-range interatomic potential. An advantage of finite-range interaction from a computational point of view is that there is a clear criterion for what atoms the ion interacts with as it travels in the target. Because of the linear term in Eq. (1.15) we call the potential with this screening function the NLHlin potential.

In Fig. 1.2 the NLHlin and the ZBL potential are shown for homonuclear atom pairs ($Z1 = Z_2$) as a function of the unscaled internuclear distance r. The ZBL screening function expectedly does not describe the drop-off towards the maximum range and overestimates the tails at least for some of the lighter elements.

It is interesting to attempt finding a screening length that compresses the screening function to a single function. Simple formulae such as Eqs. (1.11) to (1.13) cannot be expected to describe the range r_{max} of the potentials because it mainly depends on the number of valence electrons of the elements. However, one can try to compress the screening func-

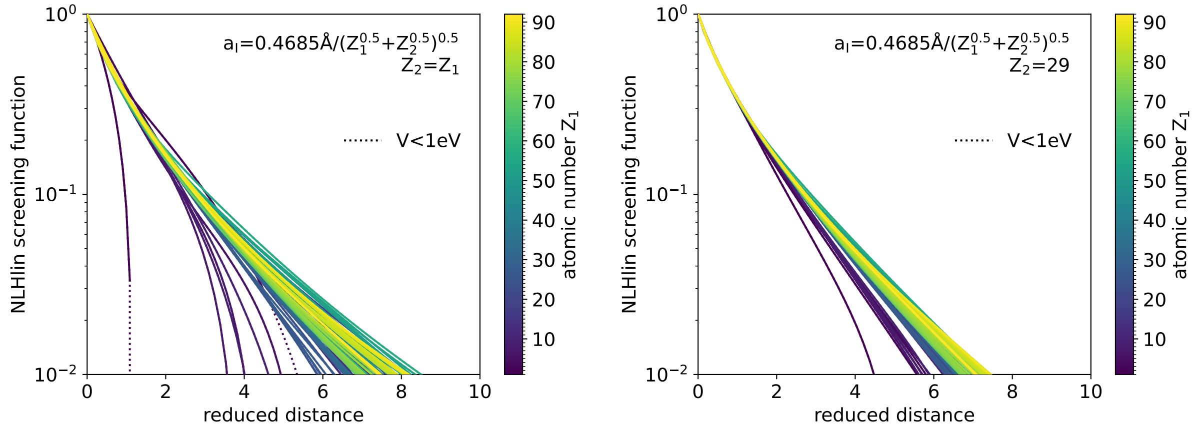

Figure 1.4: NLHlin screening function as a function of internuclear distance scaled with the proposed screening lengths Eq. (1.16) for the homo-nuclear case (left) and for Cu target atoms ($Z_2 = 29$, right).

tions at small separations, where the potential mainly depends on the core electrons. Figure 1.3 shows the NLHlin screening function $\Phi(R)$ as a function of the reduced distance $R = r/a_{\mathrm{F}}$ (left) and $R = r/a_{\mathrm{ZBL}}$ (right) using the Firsov and ZBL screening length, respectively, for homo-nuclear atom pairs. As can be seen by comparing the two plots, the ZBL screening length does a better job of compressing the functions than the Firsov screening length. Using the Firsov screening length maps the screening functions of heavy elements to larger values than those for light elements, and leaves a larger straggle in the screening functions. Also shown in Fig. 1.3 are the KrC and ZBL screening functions. To some degree, they represent an average of the individual screening functions using the respective screening lengths.

A slight improvement can be obtained by changing the power from 0.23 to 0.25 (not shown). It turns out that using screening lengths of the form

$$a_{\mathrm{I}} = \frac{0.46850\text{Å}}{(Z_1^x + Z_2^x)^y}$$

yields similar compressions as long as xy remains constant. Comparing different combination of x and y with $xy = 0.25$ indicates a slight advantage for

$$a_{\mathrm{NLH}} = \frac{0.46850\text{Å}}{\left(Z_1^{1/2} + Z_2^{1/2}\right)^{1/2}} , \qquad (1.16)$$

inparticular for very dislike elements (such as $Z_1 = 1$ or $Z_1 = 92$ combined with all other Z_2). The NLHlin screening functions as a function of reduced internuclear distance $R = r/a_{\mathrm{NLH}}$ are shown in Fig. 1.4 for the homo-nuclear case (left) and for Cu targets and all ions (right).

1.2.2 Scattering Quantities in Terms of the Screening Function

For a screened Coulomb potential, the equation for the distance of closest approach R_0 can be written in terms of reduced lengths

$$P = p/a_\mathrm{I}, \quad R = r/a_\mathrm{I}, \quad R_0 = r_0/a_\mathrm{I}, \quad T = \tau/a_\mathrm{I}\ , \tag{1.17}$$

reduced energy

$$\varepsilon = E \bigg/ \left(\frac{M_1 + M_2}{M_2} V_\mathrm{coul}(a_\mathrm{I})\right) \tag{1.18}$$

and the screening function $\Phi(R)$ as

$$1 - \frac{1}{\varepsilon R_0}\Phi(R_0) - \frac{P^2}{R_0^2} = 0\ . \tag{1.19}$$

The scattering angle in the CM system θ, Eq. (1.3), and the time integral τ, Eq. (1.8), can be written as

$$\pi - \theta(\varepsilon, P) = 2\,P \int_{R_0}^{\infty} \frac{\mathrm{d}R}{R^2\, g(R)} \tag{1.20}$$

and

$$T(\varepsilon, P) = R_0 - \int_{R_0}^{\infty} \left[\frac{1}{g(R)} - 1\right] \mathrm{d}R. \tag{1.21}$$

with

$$g(R) = \sqrt{1 - \frac{1}{\varepsilon R}\Phi(R) - \frac{P^2}{R^2}}. \tag{1.22}$$

The advantage of introducing reduced parameters lies in the fact that given a universal screening function $\Phi(R)$, Eqs. (1.19) to (1.22) depend only via Eqs. (1.17) and (1.18) on the atom species. This may be used in the development and testing of numerical algorithms, and to draw general conclusions in terms of the reduced impact parameter P and energy ε. We therefore use Eqs. (1.19) to (1.22) in the following, although formulations in terms of actual lengths (in Å) can simply be obtained by setting $a_\mathrm{I} = 1$Å and adjusting the coefficients of Φ.

1.3 Calculation of the Apsis of Collision

The apsis of collision is the distance of closest approach during a collision. The reduced apsis of collision R_0 is the solution of Eq. (1.19). For actually solving this equation with Newtons method, it is convenient to define the function $f(R) = R\, g^2(R)$ so that R_0 is the solution of:

$$f(R) = R - \frac{1}{\varepsilon}\Phi(R) - \frac{P^2}{R} = 0. \tag{1.23}$$

For screening functions given by a sum of exponentials, Eq. (1.14), and also for the finite-range screening function Eq. (1.15), the first derivative is always negative, $\Phi'(R) < 0$, and the second derivative is always positive, $\Phi''(R) > 0$. It follows that $f'(R) > 0$ and $f''(R) < 0$ for all values of R. As a consequence, Newton's method is guaranteed to converge if the initial condition $R_{0,0}$ is chosen to the left of the solution R_0 ($R_{0,0} < R_0$).

Obtaining an initial condition

For purely repulsive potentials $R_0 > P$. Therefore, P is to the left of R_0, and

$$R_0^{(0,\mathrm{P})} = P \tag{1.24}$$

is a possible initial condition, guaranteeing convergence of Newton's method. It turns out that convergence with initial condition Eq. (1.24) is slow for large scattering angles (i.e., small impact parameters P). Therefore, a second lower boundary to R_0 is desirable.

For a head-on collision ($P = 0$), Eq. (1.23) reduces to

$$R_0^{(0,\varepsilon)} - \frac{1}{\varepsilon}\Phi(R_0^{(0,\varepsilon)}) = 0 \ . \tag{1.25}$$

where we have replaced R by $R_0^{(0,\varepsilon)}$, since we want to use this equation to obtain an alternative initial condition for R_0. This equation also can be solved by iteration only. However, it is easy to produce a table $\varepsilon = \varepsilon(R_0^{(0,\varepsilon)})$ by calculating ε from given values of $R_0^{(0,\varepsilon)}$, and to obtain a value of $R_0^{(0,\varepsilon)}$ for a given ε by interpolation in the inverse table $R_0^{(0,\varepsilon)}(\varepsilon)$. If the interpolation is done cubic, the derivatives $\mathrm{d}R_0^{(0,\varepsilon)}/\mathrm{d}\varepsilon$ need to be tabulated as well. They are obtained by differentiating Eq. (1.25) with respect to ε and reinserting ε from Eq. (1.25):

$$\frac{\mathrm{d}R_0^{(0,\varepsilon)}}{\mathrm{d}\varepsilon} = -\frac{(R_0^{(0,\varepsilon)})^2}{\Phi(R_0^{(0,\varepsilon)}) - R_0^{(0,\varepsilon)}\,\Phi'(R_0^{(0,\varepsilon)})} \ . \tag{1.26}$$

Note that $\Phi'(R) < 0$ and therefore $\mathrm{d}R_0^{(0,\varepsilon)}/\mathrm{d}\varepsilon < 0$ for all $R_0^{(0,\varepsilon)}$.[6]

To set up the table, it seems more convenient to select values for ε than for $R_0^{(0,\varepsilon)}$, since the range of ε required is known to be between the implant and the trajectory cutoff energy. It is also possible to choose the ε limits generously, e.g. by $\varepsilon_{\min} = 10^{-8}$ and $\varepsilon_{\max} = 10^4$, which then would allow to use a precomputed table for all simulations.

If we desire to determine $R_0^{(0,\varepsilon)}(\varepsilon_i)$ for $\varepsilon_i = \varepsilon_{\max}/2^i$, $i = 0 \ldots n$, and $\varepsilon_{\max}/2^n \le \varepsilon_{\min} < \varepsilon_{\max}/2^{n-1}$, we can start with determining the $R_0^{(0,\varepsilon)}$ value corresponding to ε_{max} by Newton iteration of Eq. (1.23) with $P = 0$. An initial condition for the iteration, which is to the left of the solution, is $R_0^{(0,\varepsilon,0)} = 1/\varepsilon_{\max}$, which follows from Eq. (1.25) together with

[6] It can be shown that $\mathrm{d}^2R_0^{(0,\varepsilon)}/\mathrm{d}\varepsilon^2 > 0$. Therefore it is advisable to use cubic rather than linear interpolation, since linear interpolation would in general overestimate $R_0^{(0,\varepsilon)}$, which could result in $R_0^{(0,\varepsilon)} > R_0$.

$\Phi(R_0^{(0,\varepsilon,0)}) \leq 1$. Once $R_0^{(0,\varepsilon)}(\varepsilon_{\max})$ has been determined, $R_0^{(0,\varepsilon)}(\varepsilon_{\max}/2)$ is determined in the same way, using $R_0^{(0,\varepsilon)}(\varepsilon_{\max})$ as initial condition, then $R_0^{(0,\varepsilon)}(\varepsilon_{\max}/4)$ using $R_0^{(0,\varepsilon)}(\varepsilon_{\max}/2)$ as initial condition and so on, until $\varepsilon_{\max}/2^n \leq \varepsilon_{\min}$.

$R_0^{(0,\varepsilon)}$ can also be taken as an initial condition for $P > 0$. This can be seen by bringing the second and third term of Eq. (1.23) to the right-hand side and writing R_0 instead of R (since the equation is fulfilled for $R = R_0$):

$$R_0 = \frac{1}{\varepsilon}\Phi(R_0) + \frac{P^2}{R_0}.$$

Similarly, we can rewrite Eq. (1.25)

$$R_0^{(0,\varepsilon)} = \frac{1}{\varepsilon}\Phi(R_0^{(0,\varepsilon)}).$$

We can easily prove that $R_0^{(0,\varepsilon)} \geq R_0$ is not possible for any $P > 0$. If $R_0^{(0,\varepsilon)} \geq R_0$ were true for some $P > 0$, because of $\mathrm{d}\Phi/\mathrm{d}R < 0$

$$R_0 = \frac{1}{\varepsilon}\Phi(R_0) + \frac{P^2}{R_0} \geq \frac{1}{\varepsilon}\Phi(R_0^{(0,\varepsilon)}) + \frac{P^2}{R_0} = R_0^{(0,\varepsilon)} + \frac{P^2}{R_0} \tag{1.27}$$

for that P. Since $P^2/R_0 > 0$, this would mean that $R_0 > R_0^{(0,\varepsilon)}$ which contradicts the assumption $R_0^{(0,\varepsilon)} \geq R_0$. Consequently, $R_0^{(0,\varepsilon)}$ must be smaller than R_0 for all $P > 0$. So when $R_0^{(0,\varepsilon)}$ is used as an initial condition for the Newton iteration, the iteration is guaranteed to converge.

Now, as we can use both $R_0^{(0,P)}$ and $R_0^{(0,\varepsilon)}$ as initial conditions, we can choose the larger one

$$R_0^{(0)} = \max(R_0^{(0,P)}, R_0^{(0,\varepsilon)}), \tag{1.28}$$

and we will do so as it is closer to the final solution R_0 than the smaller one. Equation (1.28) can then be used as the initial condition in a Newton iteration

$$R_0^{(i+1)} = R_0^{(i)} - f(R_0^{(i)})/f'(R_0^{(i)}) \tag{1.29}$$

It guarantees convergence of the iterations for screening functions with $\Phi'(R) < 0$ and $\Phi''(r) > 0$.

Convergence

Figure 1.5 shows the number of iterations required to reach a relative accuracy of 10^{-3} between the last two calculated values of R_0 for the universal ZBL and the NLHlin potential. Usually this means that the last value is accurate to 10^{-6}, assuming quadratic convergence has been reached.

For combination of low energies and small impact parameters as well as for combinations of high energies and large impact parameters only 1-2 iterations are necessary. This is because in these cases either $R_0^{(0,P)}$ or $R_0^{(0,\varepsilon)}$ already is a good approximation to R_0. In other combination of ε and P up to 4 iterations are necessary.

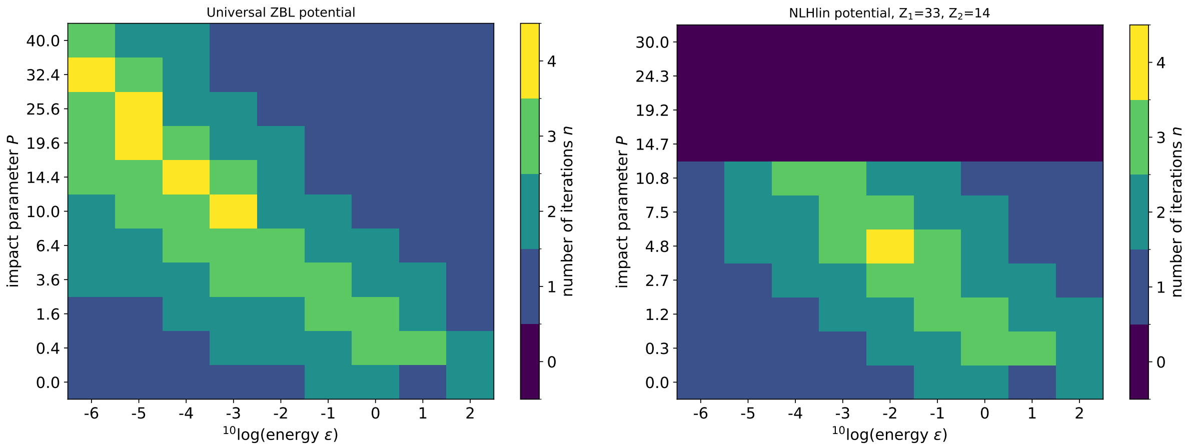

Figure 1.5: Number of iterations required to reach a relative accuracy of 10^{-3} in R_0 as a function of energy and impact parameter. Left: Universal ZBL potential. Right: NLHlin potential for As-Si using Eq. (1.16) as the screening length. The reduced impact parameters P have been chosen as to cover the same range of unscaled impact parameters p.

1.4 Calculation of the Scattering Angle in the CM System

Change of variables

The integral in Eq. (1.20) is of the form

$$\int_{R_0}^{\infty} F(R)\,\mathrm{d}R\;, \tag{1.30}$$

where R_0 is a pole of $F(R)$, which is obtained by solving Eq. (1.19), see Section 1.3. Having a numerically determined pole of the integrand as an integration boundary is inconvenient, so the first step of calculating the scattering integrals is to change the integration variable. One possibility is [1]

$$R = \frac{R_0}{(1-u^2)}\;, \tag{1.31}$$

which yields

$$\int_{R_0}^{\infty} F(R)\,\mathrm{d}R = 2R_0 \int_0^1 F\left(\frac{R_0}{1-u^2}\right)\frac{u\,\mathrm{d}u}{(1-u^2)^2}\;. \tag{1.32}$$

With Eq. (1.20)

$$\pi - \theta = 2\,P \int_{R_0}^{\infty} \frac{\mathrm{d}R}{R^2\,g(R)}$$

we obtain

$$\pi - \theta = \frac{4P}{R_0}\int_0^1 \frac{u\,\mathrm{d}u}{g\left(\frac{R_0}{1-u^2}\right)}\;, \tag{1.33}$$

where $g(R)$ is given by Eq. (1.22)

$$g(R) = \sqrt{1 - \frac{1}{\varepsilon R}\Phi(R) - \frac{P^2}{R^2}} \ .$$

In the following we show that the integrand is well-behaved in the interval $[0, 1]$.

Rewriting the integrand

It is still tricky to evaluate the integrand of Eq. (1.33) near $u = 0$ (corresponding to $R \approx R_0$), since the evaluation of $g\left(\frac{R_0}{1-u^2}\right)$ heavily depends on the accuracy of the numerically determined value of R_0. To avoid potential problems arising from this we subtract Eq. (1.19)

$$1 - \frac{1}{\varepsilon R_0}\Phi(R_0) - \frac{P^2}{R_0^2} = 0$$

from the expression under the square root of

$$g\left(\frac{R_0}{1-u^2}\right) = \sqrt{1 - \frac{1}{\varepsilon R_0}\Phi\left(\frac{R_0}{1-u^2}\right)(1-u^2) - \frac{P^2}{R_0^2}(1-u^2)^2} \ , \tag{1.34}$$

which yields

$$g\left(\frac{R_0}{1-u^2}\right) = \sqrt{\frac{R_0}{\varepsilon P^2}\,\frac{\Phi(R_0) - \Phi\left(\frac{R_0}{1-u^2}\right)(1-u^2)}{u^2} + (2-u^2)} \ . \tag{1.35}$$

Equation (1.35) can be rewritten

$$g\left(\frac{R_0}{1-u^2}\right) = u\frac{P}{R_0}\,G(u) \tag{1.36}$$

with

$$G(u) = \sqrt{\rho\,\chi(u) + (2-u^2)} \ , \tag{1.37}$$

$$\chi(u) = \frac{\Phi(R_0) - \Phi\left(\frac{R_0}{1-u^2}\right)(1-u^2)}{u^2} \ , \tag{1.38}$$

and

$$\rho = \frac{R_0}{\varepsilon P^2} \ . \tag{1.39}$$

$\chi(u)$ is well-behaved with

$$\lim_{u\to 0}\chi(u) = \Phi(R_0) - R_0\Phi'(R_0), \qquad \chi(1) = \Phi(R_0) \ . \tag{1.40}$$

The former is obtained by l'Hospital's rule. Rounding errors can be minimized by replacing Eq. (1.38) by Eq. (1.40) for u smaller than the square root of the floating point precision.

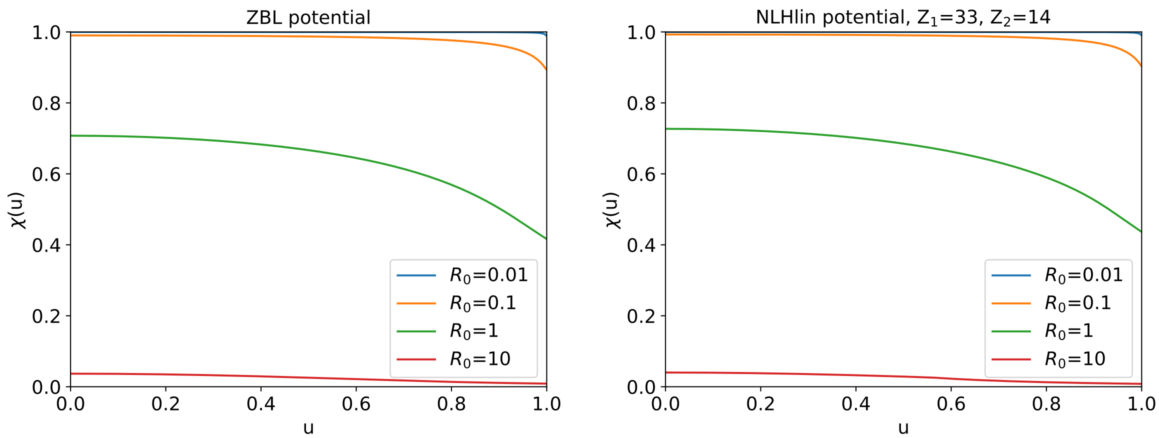

Figure 1.6: Function $\chi(u)$ according to Eq. (1.38) for the ZBL universal potential (left) and the NLHlin potential for the atom pair As-Si (right).

It is also easily seen that $\chi(u)$ is always positive for repulsive potentials ($\Phi'(R) < 0$ for $R \geq R_0$). $\chi(u)$ is plotted in Fig. 1.6 for some values of R_0.

Equation (1.33) may now be rewritten

$$\pi - \theta = 4 \int_0^1 \frac{1}{G(u)} \, \mathrm{d}u. \tag{1.41}$$

The integrand has no poles, and the integral is amenable to standard methods of numerical quadrature.

$G(u)$ evaluates to infinity for $P = 0$. To avoid handling infinite values, we define

$$H(u) = P\, G(u) = \sqrt{\frac{R_0}{\varepsilon} \chi(u) + P^2(2 - u^2)}\,. \tag{1.42}$$

Then Eq. (1.41) can be rewritten

$$\pi - \theta = 4P \int_0^1 \frac{1}{H(u)} \, \mathrm{d}u. \tag{1.43}$$

Apart from showing $\theta = \pi$, this explicitly shows that $\pi - \theta \sim P$ for small impact parameters.

Integration using Gauss-Legendre quadrature

Using Gauss-Legendre quadrature, we can approximate Eq. (1.41) by

$$\pi - \theta \approx 4 \sum_{i=1}^{n} \frac{\tilde{w}_i}{G(\tilde{u}_i)} \tag{1.44}$$

with $\tilde{u}_i$ and $\tilde{w}_i$ from Eq. (1.A.3). With $a = 0$ and $b = 1$ they read

$$\tilde{u}_i = \frac{u_i + 1}{2}, \qquad \tilde{w} = \frac{w_i}{2}. \tag{1.45}$$

Equation (1.44), however, has the problem that θ does not tend to zero exactly as $P \to \infty$ or $\varepsilon \to \infty$, since the right-hand side only approximates π. Note that for $\rho \to 0$ the integral in Eq. (1.41) becomes

$$\int_0^1 \frac{1}{\sqrt{2 - u^2}}\, \mathrm{d}u = \frac{\pi}{4}\ , \tag{1.46}$$

and $\theta \to 0$ as it should. Since the integrand is not a polynomial, the Legendre approximation of the integral will not exactly evaluate to $\pi/4$, and Eq. (1.44) will not tend to π as $\rho \to 0$. This may be fixed by scaling the integral to the correct value of $\pi/4$:

$$\pi - \theta \approx \frac{\pi}{\pi_{\mathrm{num}}/4} \sum_{i=1}^{n} \frac{\tilde{w}_i}{G(\tilde{u}_i)} \tag{1.47}$$

with

$$\frac{\pi_{\mathrm{num}}}{4} = \sum_{i=1}^{n} \frac{\tilde{w}_i}{\sqrt{2 - \tilde{u}_i^2}} \tag{1.48}$$

Since the sum in Eq. (1.47) evaluates to $\pi_{\mathrm{num}}/4$ for $\rho \to 0$, $\pi - \theta$ tends to π and $\theta \to 0$. At the same time, for $\rho \to \infty$ (i.e., for $\varepsilon \to 0$ or $P \to 0$), $G(u) \to \infty$, and the sum in Eq. (1.47) goes to zero and $\theta \to \pi$ as it should.

In terms of the function $H(u)$, Eq. (1.42), Eq. (1.47) reads

$$\pi - \theta \approx \frac{\pi}{\pi_{\mathrm{num}}/4}\, P \sum_{i=1}^{n} \frac{\tilde{w}_i}{H(\tilde{u}_i)} \tag{1.49}$$

One can also apply this correction to the Gauss-Legendre approximation of Eq. (1.33):

$$\pi - \theta \approx \frac{\pi}{\pi_{\mathrm{num}}/4} \frac{P}{R_0} \sum_{i=1}^{n} \frac{\tilde{w}_i \tilde{u}_i}{g\left(\frac{R_0}{1 - \tilde{u}_i^2}\right)}\ .$$

This expression is not numerically equivalent to Eq. (1.47). It does not require the evaluation of $\Phi(R_0)$, which might represent a noticable saving of computation time if few abscissas are used. For few abscissas, the results are very similar. With increasing number of abscissas, the results are increasingly dependent on the accuracy of the apsis R_0. The term under the square root of $g(\frac{R_0}{1-u^2})$, Eq. (1.35), or $G(u)$, Eq. (1.37), may even become negative, if the accuracy of R_0 is low and the number of abscissas is large.

Modifications for finite-range screened Coulomb potentials

If $\Phi(R) = 0$ for R larger than some R_{max},

$$\Phi(R) = 0 \quad \text{for} \quad R \geq R_{\text{max}} \tag{1.50}$$

then for $P \geq R_{\text{max}}$ we trivially have $\theta = 0$. Otherwise the integral in Eq. (1.20) must be divided in two parts:

$$\pi - \theta = 2\,P \int_{R_0}^{\infty} \frac{\mathrm{d}R}{R^2\,g(R)} = 2\,P \int_{R_0}^{R_{\text{max}}} \frac{\mathrm{d}R}{R^2\,g(R)} + 2\,P \int_{R_{\text{max}}}^{\infty} \frac{\mathrm{d}R}{R^2\,g(R)} \ . \tag{1.51}$$

The second term on the right-hand side may be evaluated analytically using $g(R)$ with $\Phi(R) = 0$

$$g(R) = \sqrt{1 - \frac{P^2}{R_0^2}} \tag{1.52}$$

and [14, 2.275.4]

$$2\,P \int_{R_{\text{max}}}^{\infty} \frac{\mathrm{d}R}{R^2\,g(R)} = 2\,P \int_{R_{\text{max}}}^{\infty} \frac{\mathrm{d}R}{R^2\sqrt{1 - \frac{P^2}{R^2}}} = 2 \arcsin \frac{P}{R_{\text{max}}} \ . \tag{1.53}$$

Note that for $R_{\text{max}} \to \infty$, this term vanishes.

Applying the substitution Eq. (1.31) to the remaining integral yields

$$\pi - \theta = 4 \int_0^{u_{\text{max}}} \frac{\mathrm{d}u}{G(u)} + 2 \arcsin \frac{P}{R_{\text{max}}} \tag{1.54}$$

with

$$u_{\text{max}} = \sqrt{1 - \frac{R_0}{R_{\text{max}}}} \ , \tag{1.55}$$

The integral can be evaluated by Gauss-Legendre quadrature

$$\pi - \theta = 4 \sum_{i=1}^{n} \frac{\tilde{w}_i}{G(\tilde{u}_i)} + 2 \arcsin \frac{P}{R_{\text{max}}} \tag{1.56}$$

with

$$\tilde{u}_i = u_{\text{max}} \frac{u_i + 1}{2}, \qquad \tilde{w}_i = u_{\text{max}} \frac{w_i}{2} \tag{1.57}$$

which follows from Eq. (1.A.3) with $a = 0$ and $b = u_{\text{max}}$.

As with Eq. (1.44), for numerical reasons $\rho \to 0$ might not exactly result in $\theta \to 0$ as it should. Unlike for infinite-range potentials, this does not happen for $P \to \infty$, because for $P > R_{\text{max}}$ we have the trivial solution $\theta = 0$. Also for $P = R_{\text{max}}$ there is no problem because the first term evaluates to zero and the second term evaluates to π. But for very high energies $\varepsilon \to \infty$ and not too small P, ρ can become very small. In that case,

$$\lim_{\varepsilon\to\infty}\left[4\int_0^{u_{\max}}\frac{\mathrm{d}u}{G(u)}\right] = 4\int_0^{u_{\max}}\frac{\mathrm{d}u}{\sqrt{2-u^2}} = 4\arcsin\frac{u_{\max}}{\sqrt{2}} = 2\arccos\frac{R_0}{R_{\max}} \tag{1.58}$$

One can therefore numerically correct Eq. (1.56) by

$$\pi - \theta = 2\,\frac{\arccos(R_0/R_{\max})}{\frac{1}{2}\arccos_{\text{num}}}\sum_{i=1}^{n}\frac{\tilde{w}_i}{G(\tilde{u}_i)} + 2\arcsin\frac{P}{R_{\max}} \tag{1.59}$$

with

$$\frac{1}{2}\arccos_{\text{num}} = \sum_{i=1}^{n}\frac{\tilde{w}_i}{\sqrt{2-\tilde{u}_i^2}}\,. \tag{1.60}$$

θ tends to 0 as $\varepsilon \to \infty$, since $\rho \to 0$ and $R_0 \to P$, and $\arccos x + \arcsin x = \pi/2$. Contrary to π_{num}, Eq. (1.48), the second sum in Eq. (1.59) depends on ε and P via $u_{\max}$ and R_0.

For $P = 0$, $\rho \to \infty$ and therefore $G(u) \to \infty$, and both terms vanish. So there is no problem in evaluating Eq. (1.59) for head-on collisions. Contrary to $\pi_{\text{num}}/4$, Eq. (1.48), $\frac{1}{2}\arccos_{\text{num}}$ depends on ε and P via $u_{\max}$ and R_0.

In terms of the function $H(u)$, Eq. (1.42), Eq. (1.59) reads

$$\pi - \theta = 2\,\frac{\arccos(R_0/R_{\max})}{\frac{1}{2}\arccos_{\text{num}}}\,P\sum_{i=1}^{n}\frac{\tilde{w}_i}{H(\tilde{u}_i)} + 2\arcsin\frac{P}{R_{\max}}\,. \tag{1.61}$$

Also for finite-range interatomic potentials, $\pi - \theta \sim P$ for small P, since $\arcsin x \approx x$ for small x.

Convergence

In Fig. 1.7, the relative errors in θ as calculated by Eq. (1.59) are shown as a function of reduced impact parameter P for reduced energies $\varepsilon = 10^{-4}$, $\varepsilon = 10^{-1}$, and $\varepsilon = 100$, and different numbers of abscissae n in the Gauss-Legendre quadrature. The results for the ZBL potential are shown in the left column, while those for the NLHlin potential are found in the right column. For the ZBL potential, the result of Biersacks magic formula [8] (check!) are also shown. Notably, the accuracy of the magic formula, about 1 %, is already reached with only $n = 2$ abscissae in the Gauss-Legendre quadrature. An accuracy 10^{-4}–10^{-3} is reached with $n = 4$. For the NLHlin potential, convergence is even faster in general.

TODO: Magic formula

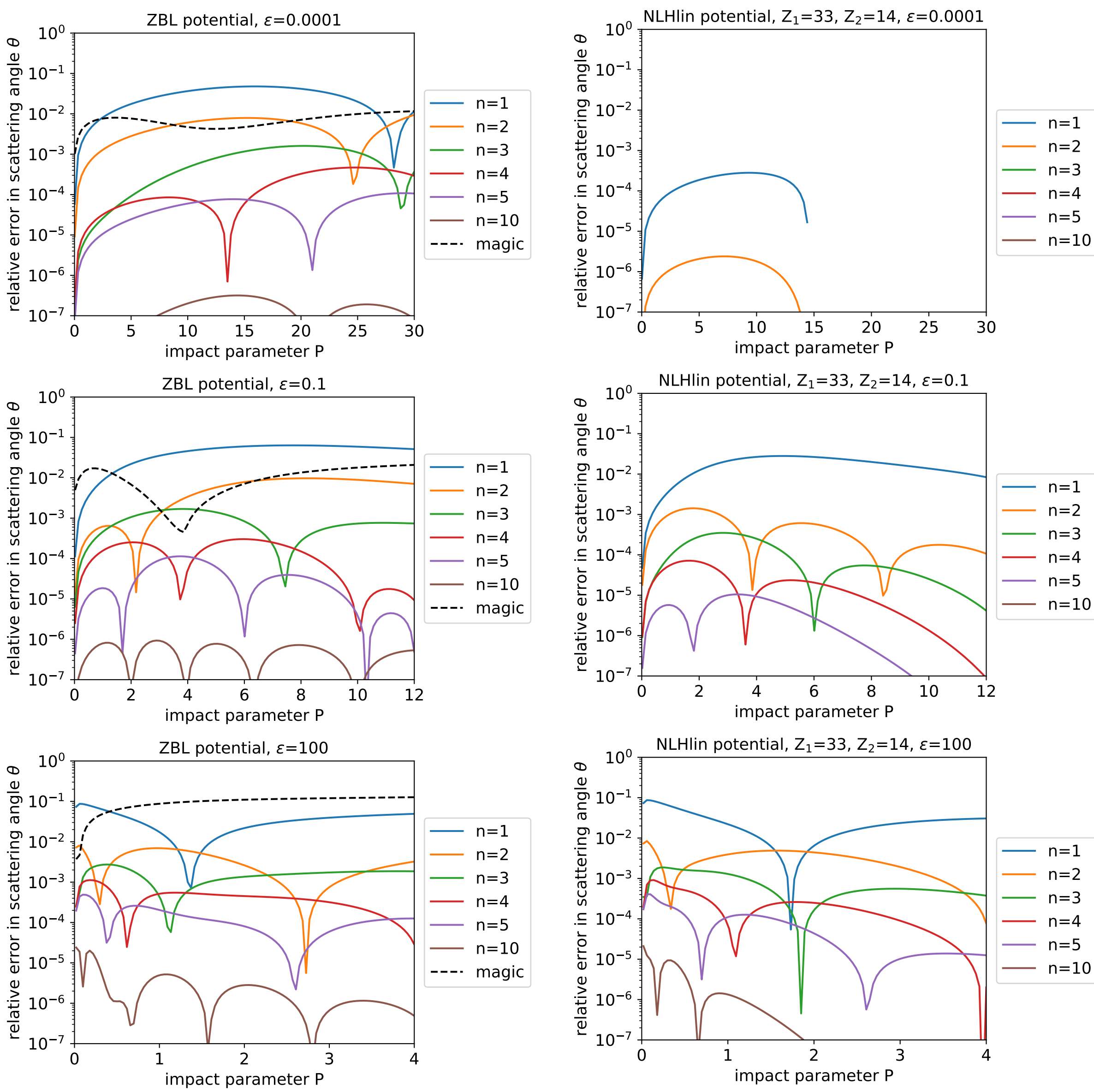

Figure 1.7: Relative errors in θ of n-point Gauss-Legendre quadrature as a function of reduced impact parameter for three energies using Eq. (1.59) with the number of abscissae n = 1,2, 3, 4, 5, and 10. As interatomic potentials the universal ZBL potential (left) and the NLHlin potential for $Z_1 = 33$ and $Z_2 = 14$ were used together with a screening length given by Eq. (1.16). For the ZBL potential the results of Biersack's magic formula [8] (check!) are also included with dashed lines. The reference simulations were evaluated using Eq. (1.59) with 32-point Gauss-Legendre quadrature. The relative accuracy of the apsis is better than 10^{-3}.

1.5 Calculation of the Time Integral

Infinite-range screened Coulomb potentials

The integral in Eq. (1.21) for the calculation of the time integral is also of the form Eq. (1.30). Applying the same substitution of variables Eq. (1.31) to Eq. (1.21), one obtains

$$T = R_0 - 2R_0 \int_0^1 \left[\frac{1}{g\left(\frac{R_0}{1-u^2}\right)} - 1 \right] \frac{u}{(1-u^2)^2}\, \mathrm{d}u \tag{1.62}$$

with $g(\frac{R_0}{1-u^2})$ as given by Eq. (1.34). Combining the two terms in brackets in Eq. (1.62) and multiplying numerator and denominator with $1 + g(\frac{R_0}{1-u^2})$ gives

$$T = R_0 - 2R_0 \int_0^1 \frac{1 - g^2\left(\frac{R_0}{1-u^2}\right)}{g\left(\frac{R_0}{1-u^2}\right)\left[1 + g\left(\frac{R_0}{1-u^2}\right)\right]} \frac{u}{(1-u^2)^2}\, \mathrm{d}u. \tag{1.63}$$

Inserting Eq. (1.34) in the numerator and Eq. (1.36) in the demoninator, and using Eqs. (1.37) to (1.39) yields

$$T = R_0 - 2P \int_0^1 \frac{1 + \rho \frac{1}{1-u^2} \Phi\left(\frac{R_0}{1-u^2}\right)}{G(u)\left[1 + u \frac{P}{R_0} G(u)\right]}\, \mathrm{d}u\ . \tag{1.64}$$

Note that the term $\frac{1}{1-u^2}\Phi\left(\frac{R_0}{1-u^2}\right)$ does not diverge if $R\,\Phi(R) \to 0$ as $R \to \infty$, in other words, if $\Phi(R)$ decays faster that $1/R$.

Using Gauss-Legendre quadrature, the time integral may be approximated by

$$T = R_0 - 2P \sum_{i-1}^{n} \tilde{w}_i \frac{1 + \rho \frac{1}{1-\tilde{u}_i^2} \Phi\left(\frac{R_0}{1-\tilde{u}_i^2}\right)}{G(\tilde{u}_i)\left[1 + \tilde{u}_i \frac{P}{R_0} G(\tilde{u}_i)\right]}\ . \tag{1.65}$$

with $\tilde{u}_i$ and $\tilde{w}_i$ defined in Eq. (1.45). In terms of the function $H(u)$, Eq. (1.42), Eq. (1.65) reads

$$T = R_0 - 2 \sum_{i-1}^{n} \tilde{w}_i \frac{P^2 + \frac{R_0}{\varepsilon} \frac{1}{1-\tilde{u}_i^2} \Phi\left(\frac{R_0}{1-\tilde{u}_i^2}\right)}{H(\tilde{u}_i)\left[1 + \frac{\tilde{u}_i}{R_0} H(\tilde{u}_i)\right]}\ . \tag{1.66}$$

which explicitly shows that the second term does not vanish for $P \to 0$.

Finite-range screened Coulomb potentials

For finite-range screened Coulomb potentials we are interested only in $P < R_{\max}$ since for $\theta = 0$ the outgoing asymptote is identical with the incoming asymptote independent of the time integral. Like for the scattering angle, we split the integral in Eq. (1.21) in two parts:

$$T = R_0 - \int_{R_0}^{R_{\max}} \left[\frac{1}{g(R)} - 1\right] \mathrm{d}R - \int_{R_{\max}}^{\infty} \left[\frac{1}{g(R)} - 1\right] \mathrm{d}R \,. \tag{1.67}$$

The second integral may be evaluated analytically using Eq. (1.52)

$$T = R_0 - (R_{\max} - \sqrt{R_{\max}^2 - P^2}) - \int_{R_{\max}}^{\infty} \left[\frac{1}{g(R)} - 1\right] \mathrm{d}R \,. \tag{1.68}$$

Applying the substitution Eq. (1.31) to the remaining integral, using the identity

$$\frac{1}{g} - 1 = \frac{1-g}{g} = \frac{1-g^2}{g(1+g)}$$

and Eqs. (1.34), (1.36), (1.37), and (1.39) we obtain

$$T = R_0 - (R_{\max} - \sqrt{R_{\max}^2 - P^2}) - 2P \int_0^{u_{\max}} \frac{1 + \rho \frac{1}{1-u^2} \Phi\left(\frac{R_0}{1-u^2}\right)}{G(u)\left[1 + u\frac{P}{R_0} G(u)\right]} \mathrm{d}u \tag{1.69}$$

with $u_{\max}$ as defined in Eq. (1.55).

Using Gauss-Legendre quadrature, the time integral may finally be approximated by

$$T = R_0 - R_{\max} + \sqrt{R_{\max}^2 - P^2} - 2P \sum_{i=1}^{n} \tilde{w}_i \frac{1 + \rho \frac{1}{1-\tilde{u}_i^2} \Phi\left(\frac{R_0}{1-\tilde{u}_i^2}\right)}{G(\tilde{u}_i)\left[1 + \tilde{u}_i \frac{P}{R_0} G(\tilde{u}_i)\right]} \,. \tag{1.70}$$

with $\tilde{u}_i$ and $\tilde{w}_i$ defined in Eq. (1.57). In terms of the function $H(u)$, Eq. (1.42), Eq. (1.70) reads

$$T = R_0 - R_{\max} + \sqrt{R_{\max}^2 - P^2} - 2 \sum_{i=1}^{n} \tilde{w}_i \frac{P^2 + \frac{R_0}{\varepsilon} \frac{1}{1-\tilde{u}_i^2} \Phi\left(\frac{R_0}{1-\tilde{u}_i^2}\right)}{H(\tilde{u}_i)\left[1 + \frac{\tilde{u}_i}{R_0} H(\tilde{u}_i)\right]} \,. \tag{1.71}$$

Convergence

In Fig. 1.8, the relative errors in $T = \tau/a_{\mathrm{I}}$ as calculated by Eq. (1.70) are shown as a function of reduced impact parameter P for reduced energies $\varepsilon = 10^{-4}$, $\varepsilon = 10^{-1}$, and $\varepsilon = 100$, and different numbers of abscissae n in the Gauss-Legendre quadrature. The results for the ZBL potential are shown in the left column, while those for the NLHlin potential are found in the right column. An accuracy of at least 1% is reached with $n = 4$. For the NLHlin potential, convergence is even faster in general.

In summary, 4-point Gauss-Legendre quadrature allows the calculation of the scattering angle with an accuracy of better than 10^{-3} and the calculation of the time integral with an accuracy of better than 1%. Using the same number of abscissae for the calculation of both quantities has the advantage that the $\Phi\left(\frac{R_0}{1-u^2}\right)$ values evaluated for the scattering angle can be reused for the time integral.

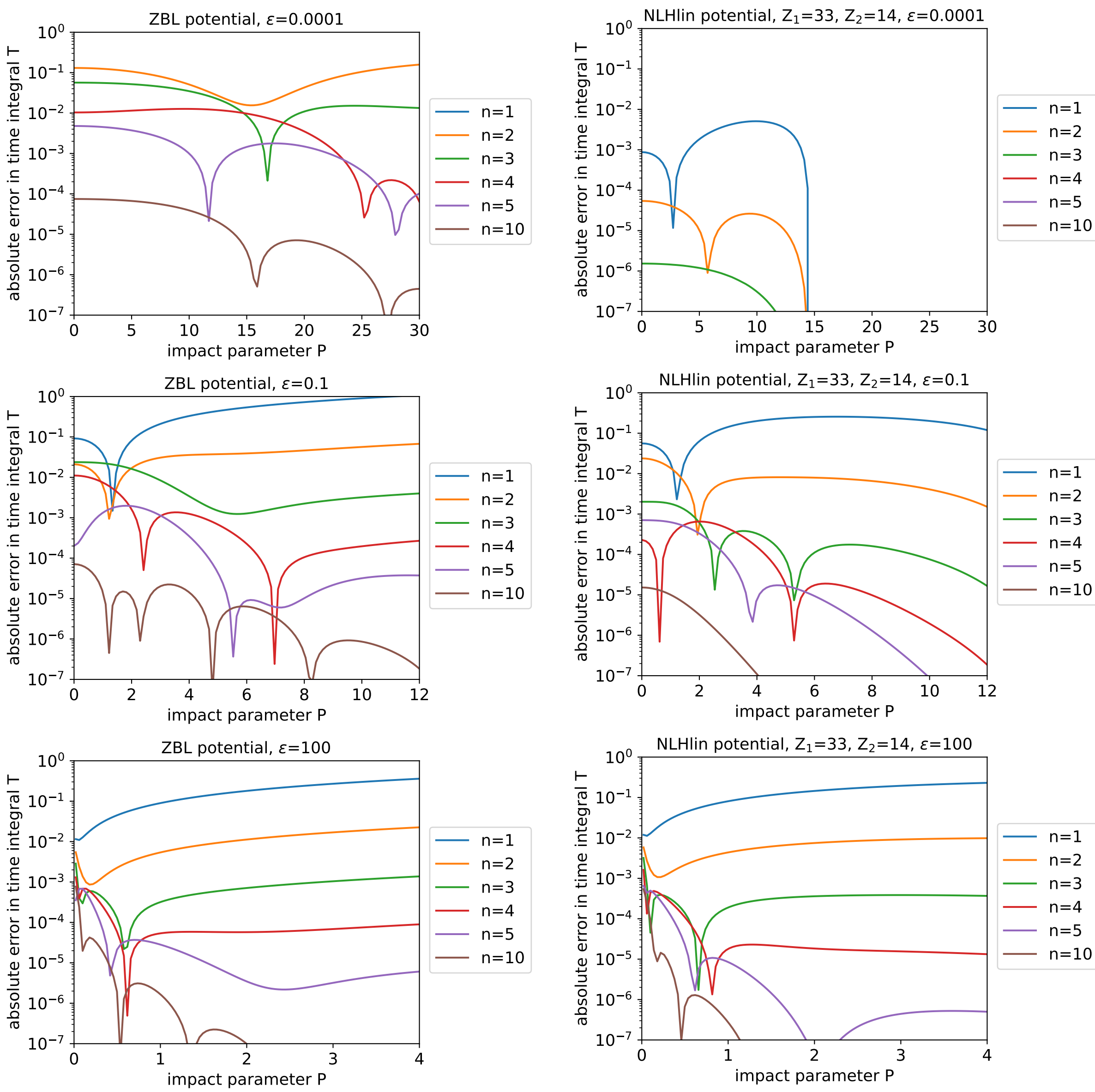

Figure 1.8: Relative errors in the time integral $T = \tau/a_\mathrm{I}$ of n-point Gauss-Legendre quadrature as a function of reduced impact parameter for three energies using Eq. (1.70) with the number of abscissae $n = 1$,2, 3, 4, 5, and 10. As interatomic potentials the universal ZBL potential (left) and the NLHlin potential for $Z_1 = 33$ and $Z_2 = 14$ were used together with a screening length a_I given by Eq. (1.16). The reference simulations were evaluated using Eq. (1.70) with 32-point Gauss-Legendre quadrature. The relative accuracy of the apsis is better than 10^{-3}.

1.6 Nuclear Stopping Cross Section and Straggling

Nuclear stopping cross section

When a projectile travels a distance $\mathrm{d}r$ in a target with randomly arranged atoms of atomic density N (units: atoms/Å^3), it passes by $2\pi p\,\mathrm{d}p\,\mathrm{d}r N$ atoms at distances between p and $p+\mathrm{d}p$. According to Eq. (1.2), in a collision with impact parameter p, the projectile transfers the energy

$$T_\mathrm{n} = \gamma\, E\, \sin^2 \frac{\theta}{2}\ , \quad \gamma = \frac{4M_1M_2}{(M_1+M_2)^2} \tag{1.72}$$

to the target atom, where $\theta = \theta(E,p)$ can be calculated as described in Section 1.4. The average total energy loss $-\overline{\mathrm{d}E}$ of the projectile to target nuclei while traveling a distance $\mathrm{d}r$ therefore is

$$-\overline{\mathrm{d}E} = \int_0^\infty T_\mathrm{n}(E,p)\, 2\pi p\, \mathrm{d}p\, \mathrm{d}r N \tag{1.73}$$

The average energy loss to nuclei per path length (also called "stopping force") therefore is

$$-\overline{\left(\frac{\mathrm{d}E}{\mathrm{d}r}\right)}_\mathrm{n} = N\, S_\mathrm{n}(E) \tag{1.74}$$

where the stopping cross section (also called "stopping power") is given by

$$S_\mathrm{n}(E) = \pi \int_0^\infty T_\mathrm{n}(E,p)\, 2p\, \mathrm{d}p = \pi\gamma E \int_0^\infty \sin^2 \frac{\theta}{2}\, 2p\, \mathrm{d}p\ . \tag{1.75}$$

In terms of the reduced quantities Eqs. (1.17) and (1.18) the stopping cross section may be written

$$S_\mathrm{n}(E) = \pi\gamma(E/\varepsilon)a_\mathrm{I}^2\, s_\mathrm{n}(\varepsilon) \tag{1.76}$$

with the reduced nuclear stopping cross section

$$s_\mathrm{n}(\varepsilon) = \varepsilon \int_0^\infty \sin^2 \frac{\theta(\varepsilon,P)}{2}\, 2P\, \mathrm{d}P \tag{1.77}$$

With Eqs. (1.18) and (1.10), Eq. (1.76) can be rewritten

$$S_\mathrm{n}(E) = \pi \frac{4M_1}{M_1+M_2} Z_1Z_2(14.4\,\mathrm{eV\AA})\, a_\mathrm{I}\, s_\mathrm{n}(\varepsilon) \tag{1.78}$$

The reduced nuclear stopping cross sections S_n for the NLHlin potentials are shown as a function of reduced energy ε in Fig. 1.9 for homo-nuclear pairs. Considerable variation is observed, in particular at low energies, which cannot be described by universal interatomic potentials like ZBL and KrC. Comparing the results for the universal and the NLHlin potentials at $\varepsilon < 1$, the ZBL potential lies at the upper limit of the NLHlin potentials, while the KrC potential better approximates the average of the heavier elements ($Z \gtrsim 40$).

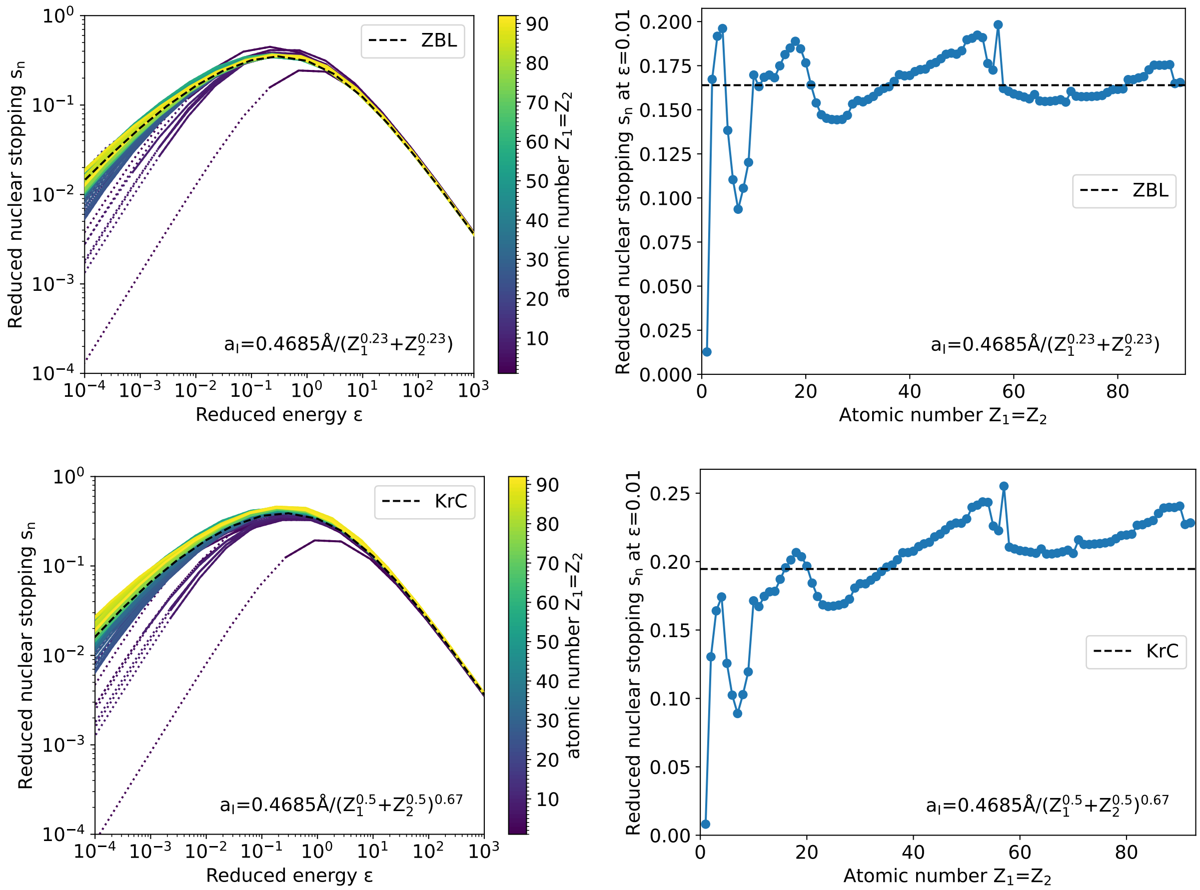

Figure 1.9: Reduced nuclear stopping cross section for the NLHlin potential (colored lines) of homonuclear atom pairs ($Z_1 = Z_2$) as a function of reduced energy ε (left) and for $\varepsilon = 0.01$ as a function of atomic number (right). Two different screening lengths have been used. Upper row: ZBL screening length Eq. (1.13); lower row: Firsov screening length Eq. (1.11) For comparison, the stopping cross sections of the ZBL potential (upper row) and the KrC potential (lower row) are depicted by black dashed lines. Note that these two potentials yield universal reduced stopping cross sections with the respective screening lengths. In the left panels the atomic numbers are color-coded. The dotted parts of the lines correspond to energies < 10 eV.

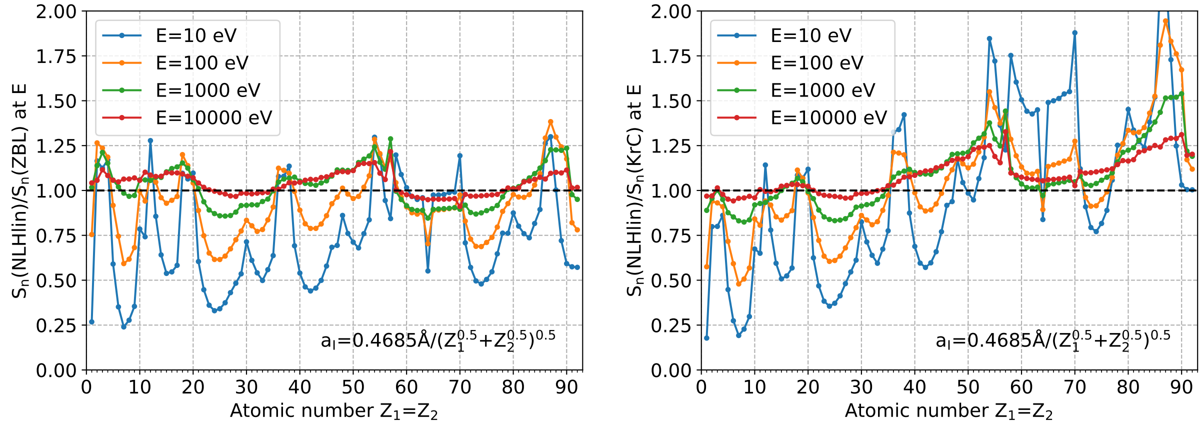

Figure 1.10: Ratio of the stopping cross section calculated using the NLHlin potential and the ZBL potential (left) or KrC potential (right) as a function of atomic number for four different energies. The data has been obtain for the homonuclear case ($Z_1 = Z_2$).

On the other hand, KrC does even worse than ZBL for the lighter elements ($Z \lesssim 20$), which is mainly due to the Firsov screening length (check!).

For the light ions the low values of the reduced energy are of no relevance since they correspond to energies where the BCA breaks down. It is therefore more instructive to plot the nuclear stopping cross section as a function of the actual energy. Fig. 1.10 shows the ratios of the stopping cross sections calculated using the NLHlin potential and the ZBL potential or KrC potential as a function of atomic number for four different energies for the homonuclear case ($Z_1 = Z_2$). Compared to when the stopping cross section is plotted for constant reduced energies (Fig. 1.9), the deviations from 1 appear more evenly distributed between light and heavy ions.

It is noted that in BCA simulations, impact parameters must be limited by some maximum value p_{max}. This value is often chosen relatively small, so that considerable energy loss is neglected at low energies (check!). Thus, the effective stopping cross section is lowered in the simulation, possibly towards the values for the NLHlin potential, however, in an unsystematic way.

The stopping cross sections for all atom pairs at $\varepsilon = 0.02$ are depicted in Fig. 1.11. The results are in line with those for the homo-nuclear pairs (Fig. 1.9).

Nuclear straggling

It may be shown [2] that the variance of the energy loss due to nuclear collisions over a path length dr in a target with randomly positioned atoms is given by

$$\overline{\left(\mathrm{d}E - \overline{\mathrm{d}E}\right)^2} = \int_0^\infty T_{\mathrm{n}}^2(E,p)\, 2\pi p\, \mathrm{d}p\, \mathrm{d}r N = N \mathrm{d}r\, Q_{\mathrm{n}}(E) \tag{1.79}$$

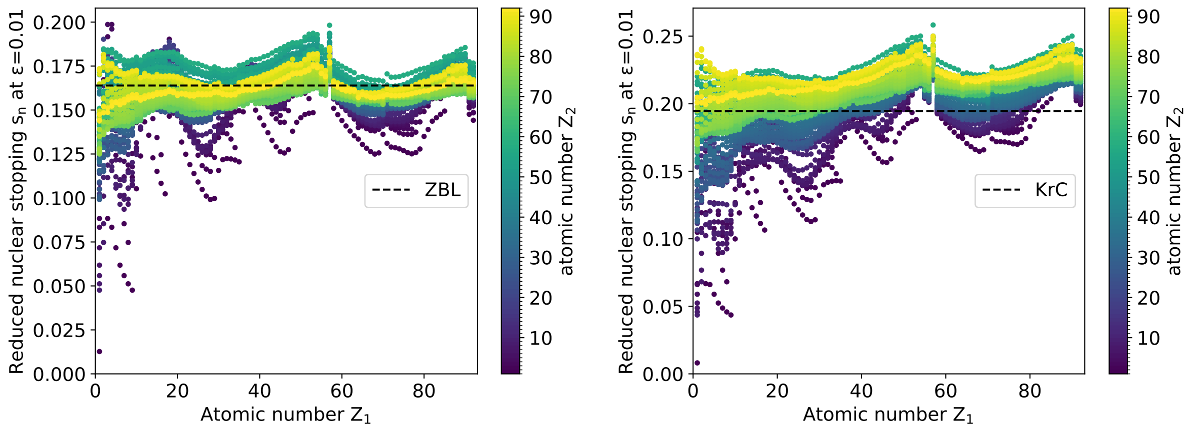

Figure 1.11: Reduced nuclear stopping cross section for the NLHlin potential of all atom pairs as a function of atomic number Z_1 at a reduced energy of $\varepsilon = 0.01$. Two different screening lengths have been used. Left: ZBL screening length Eq. (1.13); right: Firsov screening length Eq. (1.11) For comparison, the stopping cross sections of the ZBL potential (left) and the KrC potential (right) are depicted by black dashed lines. Note that these two potentials yield universal reduced stopping cross sections with the respective screening lengths. The atomic numbers Z_2 are color-coded.

with

$$Q_\mathrm{n}(E) = \int_0^\infty T_\mathrm{n}^2(E,p)\, 2\pi p \,\mathrm{d}p \;, \tag{1.80}$$

which is usually called “nuclear straggling”. Using Eq. (1.72), we obtain

$$Q_\mathrm{n}(E) = \pi\gamma^2 E^2 \int_0^\infty \sin^4\frac{\theta}{2}\, 2p\,\mathrm{d}p \;. \tag{1.81}$$

In terms of the reduced quantities Eqs. (1.17) and (1.18) the nuclear straggling may also be written

$$Q_\mathrm{n}(E) = \pi\gamma^2 (E/\varepsilon)^2 a_\mathrm{I}^2\, q_\mathrm{n}(\varepsilon) \tag{1.82}$$

with

$$q_\mathrm{n}(\varepsilon) = \varepsilon^2 \int_0^\infty \sin^4\frac{\theta(\varepsilon,P)}{2}\, 2P\,\mathrm{d}P \;. \tag{1.83}$$

With Eqs. (1.18) and (1.10), Eq. (1.82) can be rewritten

$$Q_\mathrm{n}(E) = \pi\left[\frac{4M_1}{M_1+M_2} Z_1 Z_2 (14.4\,\mathrm{eV\AA})\right]^2 q_\mathrm{n}(\varepsilon) \;. \tag{1.84}$$

The results for the nuclear straggling for the NLHlin potentials are shown in Fig. 1.12. They compare similarly to the ZBL and KrC potential as the nuclear stopping cross sections.

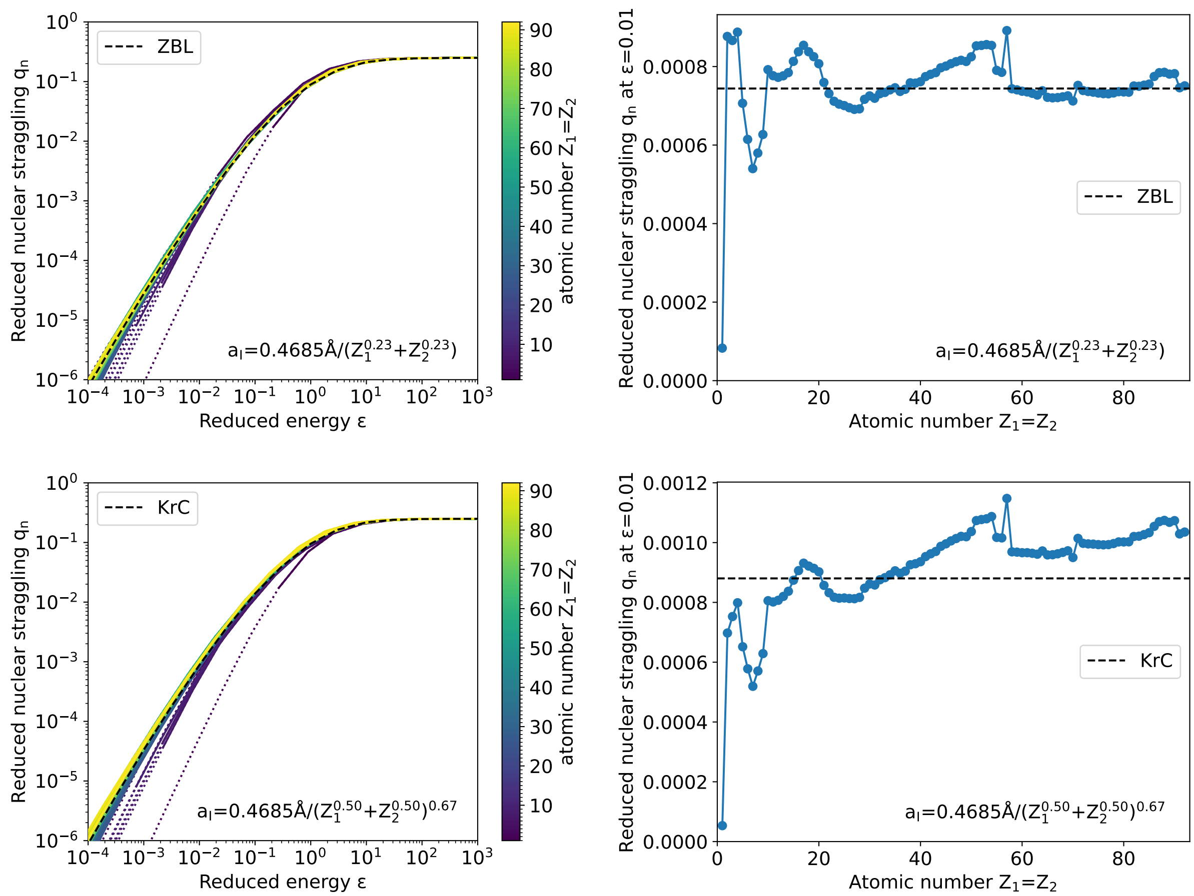

Figure 1.12: Reduced nuclear straggling for the NLHlin potential (colored lines) of homonuclear atom pairs ($Z_1 = Z_2$) as a function of reduced energy ε (left) and for $\varepsilon = 0.01$ as a function of atomic number (right). Two different screening lengths have been used. Upper row: ZBL screening length Eq. (1.13); lower row: Firsov screening length Eq. (1.11) For comparison, the stopping cross sections of the ZBL potential (upper row) and the KrC potential (lower row) are depicted by black dashed lines. Note that these two potentials yield universal reduced stragglings with the respective screening lengths. In the left panels the atomic numbers are color-coded. The dotted parts of the lines correspond to energies < 10 eV.

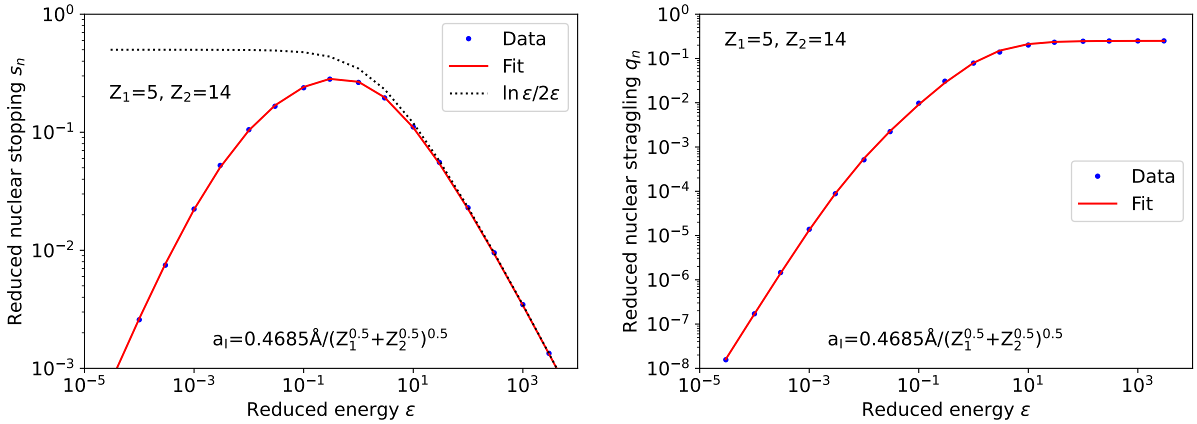

Figure 1.13: Data and fits for $Z_1 = 5$, $Z_2 = 14$.

Analytic expressions

The results for the nuclear stopping cross section and straggling can be fitted by analytical expressions within a few percent as proposed in [3]. For the stopping cross section we use

$$s_\mathrm{n}(\varepsilon) = \frac{\ln(1 + a\,\varepsilon)}{2(\varepsilon + b\,\varepsilon^c + d\,\varepsilon^{0.5})} \tag{1.85}$$

In [3] it is suggested to use $s_\mathrm{n} = \ln\varepsilon/2\varepsilon$ for $\varepsilon > 30$, which is referred to as "unscreened nuclear stopping" without further explanation. We did not find it necessary to apply this special treatment for high energies, and found actually better fitting of the data using the expression Eq. (1.85) over the whole energy range (at least) up to $\varepsilon = 10^4$.

For the nuclear straggling we use [3]

$$q_\mathrm{n}(\varepsilon) = \frac{1}{4 + a\,\varepsilon^b + c\,\varepsilon^d} \tag{1.86}$$

The values a, b, c, and d have been fitted to energies between $\varepsilon = 3 \times 10^{-5}$ and 10^4 for both nuclear stopping cross section and nuclear straggling for all atom pairs.

Figure 1.13 shows, as an example the data and fits for $Z_1 = 5$, $Z_2 = 14$.

1.7 Range

The range of the ions is determined by both nuclear and electronic stopping. The average energy loss per path length is therefore given by

$$\frac{\mathrm{d}E'}{\mathrm{d}r} = \left(\frac{\mathrm{d}E'}{\mathrm{d}r}\right)_\mathrm{n} + \left(\frac{\mathrm{d}E'}{\mathrm{d}r}\right)_\mathrm{e} = -N(S_\mathrm{n}(E') + S_\mathrm{e}(E')) \tag{1.87}$$

where S_e characterizes the energy loss to electrons but is otherwise defined like S_n. Compared to Eq. (1.74), we have replaced energy E by E' to distinguish the initial energy E

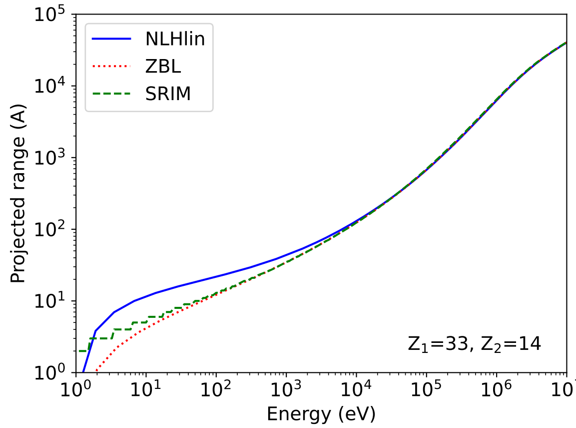

Figure 1.14: Projected range R_p as a function of energy for the NLHlin potential (solid line) and the ZBL potential (dashed and dotted lines). For the ZBL potential, our own results (dotted line) agree well with those obtained by SRIM [11].

from the current energy E'. Separation of variables and integration of the path length from 0 to R and of energy from E to 0 gives the total path length

$$R = \int_0^E \frac{\mathrm{d}E'}{N(S_\mathrm{n}(E') + S_\mathrm{e}(E'))} \tag{1.88}$$

In differential form this reads

$$\frac{\mathrm{d}R}{\mathrm{d}E} = \frac{1}{N(S_\mathrm{n}(E) + S_\mathrm{e}(E))} \tag{1.89}$$

We have silently neglected the statistical nature of nuclear stopping by dropping the overbar of Eq. (1.74). Lindhard [15] showed that R is a satisfactory approximation of the mean range along the path.

Projected range is defined by the projection of the stop position of an ion to its initial direction, measured from its initial position. The distribution of projected ranges is what is measured when a depth profile is determined of ions implanted into a flat target at normal incidence, with its origin at the surface. The most basic characteristics of this distribution is the mean penetration depth or mean projected range R_p, which in sloppy parlance is called "the" projected range.

It has been shown [16, 17], that an approximation to the projected range can be calculated from

$$\frac{\mathrm{d}R_\mathrm{p}}{\mathrm{d}E} = \frac{1}{N(S_\mathrm{n}(E) + S_\mathrm{e}(E))}\left(1 - \frac{\mu N S_\mathrm{n}(E)}{2E} R_\mathrm{p}\right), \quad \mu = \frac{M_2}{M_1} \tag{1.90}$$

Using tabulated S_e values from SRIM [11], the projected ranges have been calculated as a function of energy. Results for As in Si are shown in Fig. 1.14. The ratios of the projected ranges obtained with the NLHlin potentials and the ZBL potential are shown in Fig. 1.15 as a function of implantation energy. The ratios of the projected

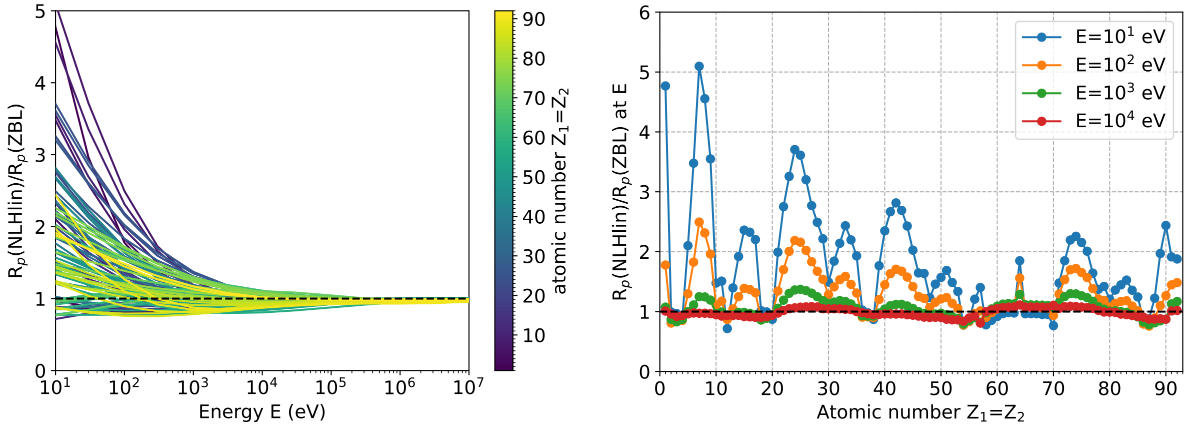

Figure 1.15: The ratio of the projected ranges calculated with the NLHlin and ZBL potentials as a function of energy (left) and as a function of atomic number (right) for homonuclear atom pairs.

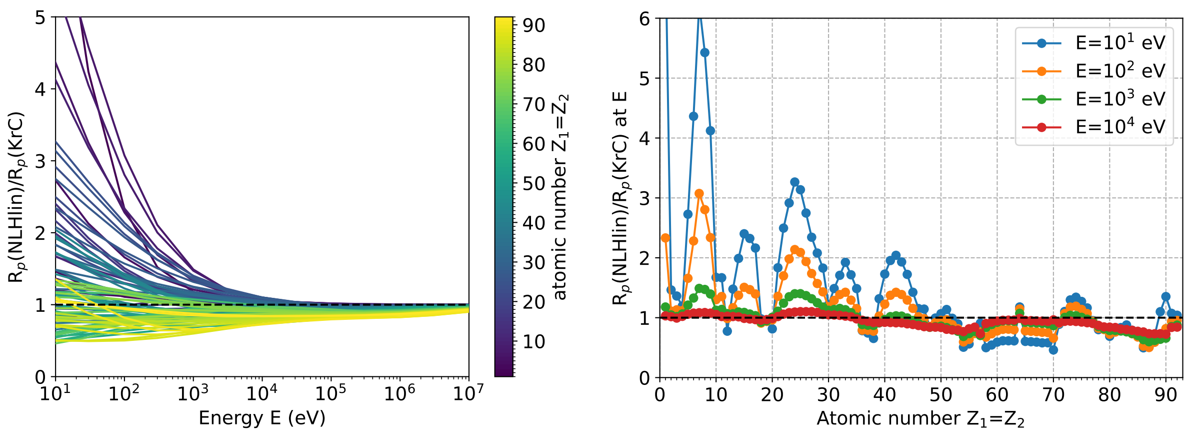

Figure 1.16: The ratio of the projected ranges calculated with the NLHlin and KrC potentials as a function of energy (left) and as a function of atomic number (right) for homonuclear atom pairs.

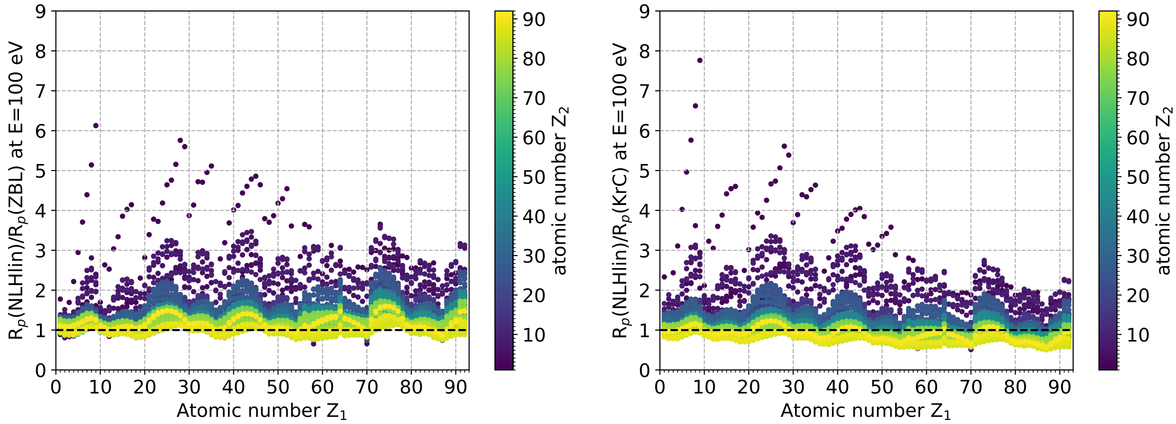

Figure 1.17: Projected range ratio with respect to ZBL (left) and KrC (right) for all combinations of Z_1 and Z_2.

ranges obtained with NLHlin and KrC potential is depicted in Fig. 1.16. Very significant deviations between the NLHlin and the conventional potentials can be seen except for the noble gases ($Z = 2, 10, 18, 36, 54, 86$). Comparing Figs. 1.15 and 1.16, KrC performs much better than ZBL for heavy atoms, while KrC performs poorer for lighter elements.

1.A Gauss-Legendre Quadrature

Gauss-Legendre quadrature approximates the integral

$$\int_{-1}^{1} h(u)\,\mathrm{d}u \approx \sum_{i=1}^{n} w_i h(u_i) \tag{1.A.1}$$

where the abscissas u_i are the roots of the Legendre polynomial $P_n(u)$ and the weights w_i are given by [18, 25.4.29]

$$w_i = \frac{2}{(1-u_i^2)[P_n'(u_i)]^2} \tag{1.A.2}$$

The roots and abscissas are usually not calculated but taken from tabulated values, e.g. [19].

Adjustment of the integration boundaries may be achieved by a linear transformation of the integration variable:

$$\int_a^b h(u)\,\mathrm{d}u = \frac{b-a}{2}\int_{-1}^{1} h\left(\frac{(b-a)u+(b+a)}{2}\right)\mathrm{d}u \approx \sum_{i=1}^{n} \tilde{w}_i\, h\left(\tilde{u}\right)$$

with

$$\tilde{u}_i = \frac{(b-a)u_i+(b+a)}{2}, \qquad \tilde{w}_i = \frac{b-a}{2} w_i\ . \tag{1.A.3}$$

Gauss-Legendre quadrature integrates polynomials of order $< 2n$ exactly, so it is most accurate if all derivatives up to order $2n-1$ exist and higher derivatives are small.

1.B Impulse approximation

For high enough energies E or small enough scattering angles, it may be assumed that the incoming projectile moves on a straight line at constant velocity. To apply this approximation, a few notes on the center-of-mass system are necessary. The transformation of the two-body system into the center-of-mass system, whose origin is in the center of mass of the projectile (atom 1) and the recoil (atom 2),

$$\vec{x}_\mathrm{c} = \frac{m_1}{m_1+m_2}\vec{x}_1 + \frac{m_2}{m_1+m_2}\vec{x}_2\,, \tag{1.B.1}$$

can be described in terms of the vector from recoil to projectile

$$\vec{x}_\mathrm{r} = \vec{x}_1 - \vec{x}_2\,. \tag{1.B.2}$$

The time evolution of $\vec{x}_\mathrm{r}$ is the same as that of the position of a single particle with the (so-called) reduced mass

$$m_\mathrm{r} = \frac{m_1 m_2}{m_1+m_2}\,. \tag{1.B.3}$$

in a stationary potential $V(\vec{x}_\mathrm{r})$. The (constant) energy of the one-particle system with the reduced mass m_r equals the sum of the energies of the two particles with masses m_1 and m_2 in the center-of-mass system. Its value is

$$E_\mathrm{r} = \frac{m_\mathrm{r} v_0^2}{2} \tag{1.B.4}$$

where v_0 is the velocity of the projectile in the laboratory system before the collision. Note that it equals the velocity $\dot{\vec{x}}_\mathrm{r} = \dot{\vec{x}}_1 - \dot{\vec{x}}_2$ of the particle in the one-particles system, since before the collision $\dot{\vec{x}}_2 = 0$ and the potential energy is zero. With the energy in the laboratory system $E = m_1 v_0^2/2$, we regain Eq. (1.5):

$$E_\mathrm{c} = \frac{M_2}{M_1 + M_2}\, E.$$

Now assume that a projectile with mass m_r moves at constant speed $v_\mathrm{x} = v_0$ parallel to the x axis at $y = p$ in a potential $V(r)$ centered at the origin. Its momentum is zero initially. During passage, the projectile receives an impulse (change in momentum) of

$$m_\mathrm{r}(v_\mathrm{y} - 0) = \int_{-\infty}^{\infty} F_\mathrm{y}(t)\,\mathrm{d}t = \frac{1}{v_\mathrm{x}} \int_{-\infty}^{\infty} F_\mathrm{y}(x)\,\mathrm{d}x \tag{1.B.5}$$

where v_y denotes the y component of the velocity after the collision and F_y the y component of the force. The scattering angle in the center-of-mass system is given by

$$\theta \approx \tan\theta = \frac{v_\mathrm{y}}{v_\mathrm{x}} = \frac{1}{2E_\mathrm{c}} \int_{-\infty}^{\infty} F_\mathrm{y}(x)\,\mathrm{d}x = \frac{1}{2E}\frac{M_1 + M_2}{M_2} \int_{-\infty}^{\infty} F_\mathrm{y}(x)\,\mathrm{d}x \tag{1.B.6}$$

where we have used $E_\mathrm{c} = m_\mathrm{r} v_\mathrm{x}^2/2$ and Eq. (1.5). According to Eqs. (1.1) and (1.2), the scattering angle in the laboratory system and the energy transfer are given by

$$\psi_1 \approx \tan\psi_1 \approx \frac{M_2}{M_1 + M_2}\,\theta \approx \frac{1}{E}\left(\frac{1}{2}\int_{-\infty}^{\infty} F_\mathrm{y}(x)\,\mathrm{d}x\right) \tag{1.B.7}$$

and

$$\Delta E \approx \frac{M_1 M_2}{(M_1 + M_2)^2}\, E\,\theta^2 \approx \frac{1}{E}\,\frac{M_1}{M_2}\left(\frac{1}{2}\int_{-\infty}^{\infty} F_\mathrm{y}(x)\,\mathrm{d}x\right)^2, \tag{1.B.8}$$

respectively. Like θ, ψ_1 and ΔE are proportional to $1/E$. Interestingly, ψ_1 does not depend on the atom masses M_1 and M_2.

For a spherically symmetric potential $V(r)$, the force in y direction is given by

$$F_\mathrm{y}(x) = -\frac{\mathrm{d}V(r)}{\mathrm{d}r}\frac{p}{r} = -\frac{\mathrm{d}V(r)}{\mathrm{d}p} \tag{1.B.9}$$

with

$$r = \sqrt{x^2 + p^2}. \tag{1.B.10}$$

The first expression after the equal sign in Eq. (1.B.9) takes the y component of the force, which points in the direction of the vector from the origin to the projectile. The second expression can be shown to be equivalent to the first one by applying the chain rule of differentiation. Using either expression and making use of $\int_{-\infty}^{0} = \int_{0}^{\infty}$, we obtain

$$\frac{1}{2}\int_{-\infty}^{\infty} F_{\mathrm{y}}(x)\,\mathrm{d}x = -\int_{0}^{\infty} \frac{\mathrm{d}V(r)}{\mathrm{d}r}\frac{p}{r}\,\mathrm{d}x = -\frac{\mathrm{d}}{\mathrm{d}p}\int_{0}^{\infty} V(r)\,\mathrm{d}x \tag{1.B.11}$$

where in the second expression we have also changed the order of integration and differentiation.

If we further assume a screened Coulomb potential, Eq. (1.9), we obtain

$$\frac{1}{2}\int_{-\infty}^{\infty} F_{\mathrm{y}}(x)\,\mathrm{d}x = Z_1 Z_2 e^2 \int_{0}^{\infty} \frac{\Phi(r) - r\,\Phi'(r)}{r^3} p\,\mathrm{d}x = -Z_1 Z_2 e^2 \frac{\mathrm{d}}{\mathrm{d}p}\int_{0}^{\infty} \frac{\Phi(r)}{r}\,\mathrm{d}x \tag{1.B.12}$$

Using reduced lengths, Eq. (1.17) and $X = x/a_{\mathrm{I}}$, Eq. (1.B.12) can be rewritten as

$$\frac{1}{2}\int_{-\infty}^{\infty} F_{\mathrm{y}}(x)\,\mathrm{d}x = \frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}} I(P) \tag{1.B.13}$$

with the dimensionless integral

$$I(P) = \int_{0}^{\infty} \frac{\Phi(R) - R\,\Phi'(R)}{R^3} P\,\mathrm{d}X = -\frac{\mathrm{d}}{\mathrm{d}P}\int_{0}^{\infty} \frac{\Phi(R)}{R}\,\mathrm{d}X. \tag{1.B.14}$$

Inserting Eq. (1.B.13) in Eqs. (1.B.7) and (1.B.8), we obtain

$$\psi_1 \approx \frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}} \frac{I(P)}{E} \tag{1.B.15}$$

and

$$\Delta E \approx \frac{M_1}{M_2}\left(\frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}}\right)^2 \frac{I(P)^2}{E}. \tag{1.B.16}$$

The second expression on the right-hand side of Eq. (1.B.14) is favorable for infinite range sum-of-exponentials screening functions, Eq. (1.14). Using the transformation $t = R/P = \sqrt{1 + X^2/P^2}$, the integrals evaluate to [14, 8.432.3]

$$\int_{0}^{\infty} \frac{\exp(-b_i R)}{R}\,\mathrm{d}X = \int_{1}^{\infty} \frac{\exp(-b_i P t)}{\sqrt{t^2 - 1}}\,\mathrm{d}t = K_0(b_i P)$$

where $K_0(x)$ is the modified Bessel function of second kind and order 0. $I(P)$ therefore evaluates to

$$I(P) = -\sum_i a_i \frac{\mathrm{d}K_0(b_i P)}{\mathrm{d}P} = \sum_i a_i b_i K_1(b_i P) \tag{1.B.17}$$

where $K_1(x)$ is the modified Bessel function of second kind and order 1. For other screening functions, the first integral on the right-hand side of Eq. (1.B.14) is preferable, since it does not require numerical differentiation.

Selection of the maximum impact parameter

The impulse approximation can be used to estimated the maximum impact parameter that is necessary to to consider all collisions with scattering angles and energy transfers about some chosen limit. This is because the desired limit of the scattering angle usually is quite low, say $\sim 1°$.

With a purely repulsive interatomic potential, increasing impact parameters P result in decreasing scattering angles ψ_1, so according to Eq. (1.B.15), $I(P)$ should be a monotonically decreasing function.[7] Then, according to Eq. (1.B.16), also $\Delta E(P)$ is monotonically decreasing. Thus, if we want to guarantee that by choosing the impact parameter between 0 and P_{max} we do not miss any collisions with either $\psi_1 > \psi_{\mathrm{min}}$ or $\Delta E > \Delta E_{\mathrm{min}}$, both $\psi_1(P_{\mathrm{max}}) \leq \psi_{\mathrm{min}}$ and $\Delta E(P_{\mathrm{max}}) \leq \Delta E_{\mathrm{min}}$ must hold. The minimum P_{max} for which both conditions are met is the one where the "$\leq$" sign is replaced by the "$=$" sign in one of the conditions, whatever occurs first when P_{max} is decreased.

As can be seen from Eqs. (1.B.15) and (1.B.16), the resulting conditions for P_{max} depend on the projectile energy E. On the other hand, for a given P_{max}, conditions for E result from $\psi_1(P_{\mathrm{max}}) \leq \psi_{\mathrm{min}}$ and $\Delta E(P_{\mathrm{max}}) \leq \Delta E_{\mathrm{min}}$:

$$E \geq \frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}} \frac{I(P_{\mathrm{max}})}{\psi_{\mathrm{min}}} \tag{1.B.18}$$

$$E \geq \frac{M_1}{M_2} \left(\frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}} \right)^2 \frac{I(P_{\mathrm{max}})^2}{\Delta E_{\mathrm{min}}} \,. \tag{1.B.19}$$

from $\Delta E(P_{\mathrm{max}}) \leq \Delta E_{\mathrm{min}}$. Since both conditions must be met, E must be larger than or equal to the larger one of the two right-hand sides of Eqs. (1.B.18) and (1.B.19). If we now denote as P_{max} the **smallest value of the maximum impact parameter that fulfills the conditions**, we have to replace the "$\geq$" sign by the "$=$" sign:

$$E = \max \left(\frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}} \frac{I(P_{\mathrm{max}})}{\psi_{\mathrm{min}}}, \frac{M_1}{M_2} \left(\frac{Z_1 Z_2 e^2}{a_{\mathrm{I}}} \right)^2 \frac{I(P_{\mathrm{max}})^2}{\Delta E_{\mathrm{min}}}. \right) . \tag{1.B.20}$$

Eq. (1.B.20) can be used to generate a table of P_{max}-E pairs, which can then be used to interpolate P_{max} values for given energies E.

The selection of the maximum impact parameter is illustrated in Fig. 1.B.1 with two examples. In the first example, W is bombarded with He at 3 keV. One is interested in sputtering, therefore the surface binding energy is used in the energy criterion. Moderately accurate trajectories are desired, so the minimum scattering angle is chosen as 3°. The two choices correspond to two maximum impact parameters. The larger one, resulting from the desired minimum scattering angle, has to be selected, since both conditions must be fulfilled. In the other example shown in Fig. 1.B.1, Si is bombarded with 100 keV As. Here one wishes to include all damage generation events in the simulation, so a minimum

[7] This could be violated when the impulse approximation fails.

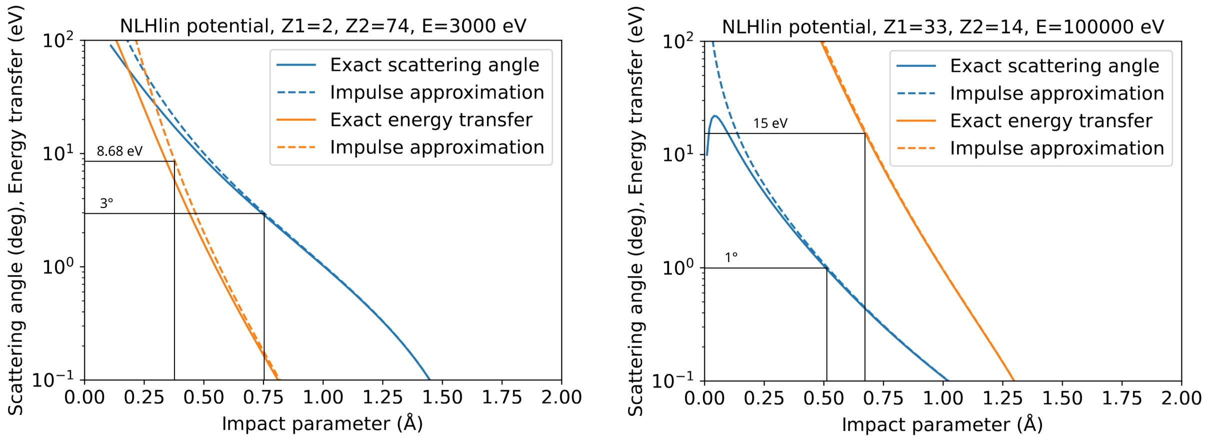

Figure 1.B.1: Determination of the maximum impact parameter from minimum scattering angle and energy transfer. Left: 3 keV He in W, using the surface binding energy of W (8.68 eV) as the minimum energy transfer and 3° as the minimum scattering angle. Right: 100 keV As in Si, using the threshold displacement energy of Si (15 eV) as the minimum energy transfer and 1° as the minimum scattering angle.

energy transfer equal to the threshold displacement energy is chosen. Accurate trajectories are warranted by a small minimum scattering angle of 1°. Again, the larger one of the two resulting maximum impact parameters must be chosen.

Note that the impulse approximation always yields upper limits to the energy transfer and scattering angle. Their graphs as a function of impact parameter therefore are always to the right of the exact values (dashed vs. solid lines in Fig. 1.B.1). The estimate of the maximum impact parameter therefore is an upper limit to what the exact calculation would yield. This means, using the maximum impact parameter so determined, one might occasionally include events with energy transfers or scattering angles below the chosen limits, but one would not omit events above these limits.

1.C Alternative Calculation of the Apsis of Collision

Obtaining an initial condition

An alternative approach to finding an initial solution of Eq. (1.23),

$$f(R) = R - \frac{1}{\varepsilon}\Phi(R) - \frac{P^2}{R} = 0. \tag{1.C.1}$$

is to approximate $\Phi(R)$ by a piecewise function $\phi(R)$ which is always larger than or equal to $\Phi(R)$, and which allows Eq. (1.C.1). Such a function is

$$\phi(R) = \begin{cases} 1 - k_1 R & \text{for } 0 \leq R \leq R_{12} \\ k_2/R & \text{for } R_{12} \leq R \leq R_{23} \\ k_3/R^3 & \text{for } R_{23} \leq R \leq R_{34} \\ k_4(R_{\max} - R) & \text{for } R_{34} \leq R \end{cases} \tag{1.C.2}$$

where the last case is used only with a finite maximum range $R_{\max}$, and R_{12}, R_{23}, and R_{34} are chosen such that the function is continuous and its first derivative is continuous at R_{12} and R_{34}:

$$k_1 = \frac{1}{4k_2}, \quad R_{12} = \sqrt{\frac{k_2}{k_1}}, \quad R_{23} = \sqrt{\frac{k_3}{k_2}}, \quad R_{34} = \frac{3}{4}R_{\max}, \quad k_4 = \frac{3k_3}{R_{34}^4} \tag{1.C.3}$$

The constants k_2 and k_3 are chosen such that the function touches $\Phi(R)$, i.e., $\phi(R) = \Phi(R)$ and $\phi'(R) = \Phi'(R)$. The touch points and coefficients are obtained by solving the equations

$$\Phi(R_2) + R_2\Phi'(R_2) = 0, \qquad k_2 = R_2\Phi(R_2) \tag{1.C.4}$$

and

$$\Phi(R_3) + \frac{1}{3}R_3\Phi'(R_3) = 0, \qquad k_3 = R_3^3\Phi(R_3) \tag{1.C.5}$$

In order to approximately solve Eq. (1.C.1), we replace $\Phi(R)$ by $\phi(R)$ for each of the three intervals and obtain

$$R_0^{(0)} = \begin{cases} \dfrac{1 + \sqrt{1 + 4(\varepsilon + k_1)\varepsilon P^2}}{2(\varepsilon + k_1)} & \text{for } 0 \leq R \leq R_{12} \\ \sqrt{P^2 + \frac{k_2}{\varepsilon}} & \text{for } R_{12} \leq R \leq R_{23} \\ \sqrt{\frac{P^2}{2} + \sqrt{\frac{P^4}{4} + \frac{k_3}{\varepsilon}}} & \text{for } R_{23} \leq R \leq R_{34} \\ \dfrac{k_4 R_{\max} + \sqrt{(k_4 R_{\max})^2 + 4(\varepsilon + k_4)\varepsilon P^2}}{2(\varepsilon + k_4)} & \text{for } R_{34} \leq R \end{cases} \tag{1.C.6}$$

Inserting the solutions $R_0^{(0)}$ into the conditions for R and using Eq. (1.C.3), one can rewrite the conditions in terms of $P^2 + k_2/\varepsilon$, so that we finally obtain

$$R_0^{(0)} = \begin{cases} \dfrac{1+\sqrt{1+4(\varepsilon+k_1)\varepsilon P^2}}{2(\varepsilon+k_1)} & \text{for } P^2 \le \frac{k_2}{k_1} - \frac{k_2}{\varepsilon} \\ \sqrt{P^2 + \frac{k_2}{\varepsilon}} & \text{for } \frac{k_2}{k_1} - \frac{k_2}{\varepsilon} \le P^2 \le \frac{k_3}{k_2} - \frac{k_2}{\varepsilon} \\ \sqrt{\frac{P^2}{2} + \sqrt{\frac{P^4}{4} + \frac{k_3}{\varepsilon}}} & \text{for } \frac{k_3}{k_2} - \frac{k_2}{\varepsilon} \le P^2 \le R_{34}^2 - \frac{k_3}{\varepsilon R_{34}^2} \\ \dfrac{k_4 R_{\max} + \sqrt{(k_4 R_{\max})^2 + 4(\varepsilon+k_4)\varepsilon P^2}}{2(\varepsilon+k_4)} & \text{for } R_{34}^2 - \frac{k_3}{\varepsilon R_{34}^2} \le P^2 \end{cases} \tag{1.C.7}$$

The case $r > R_{34}$ cannot be used for screening functions with infinite range. Instead, the procedure implemented in TRIM85 [3] can be used, which solves the equation Eq. (1.C.1) for the head-on case ($P = 0$) for an approximate screening function $\Phi(R) = \exp(-0.37R)$ by fix-point iteration:

$$R = \frac{1}{\varepsilon}\exp(-0.37R) \tag{1.C.8}$$

Starting with $R = P$, up to two iterations are done, and the algorithm falls back to $R = P$ if R gets smaller than P during iteration. If the approximation to the apsis determined in this way is smaller than the touch point of k_3/R^3, the remaining cases of Eq. (1.C.7) can be used.

Convergence

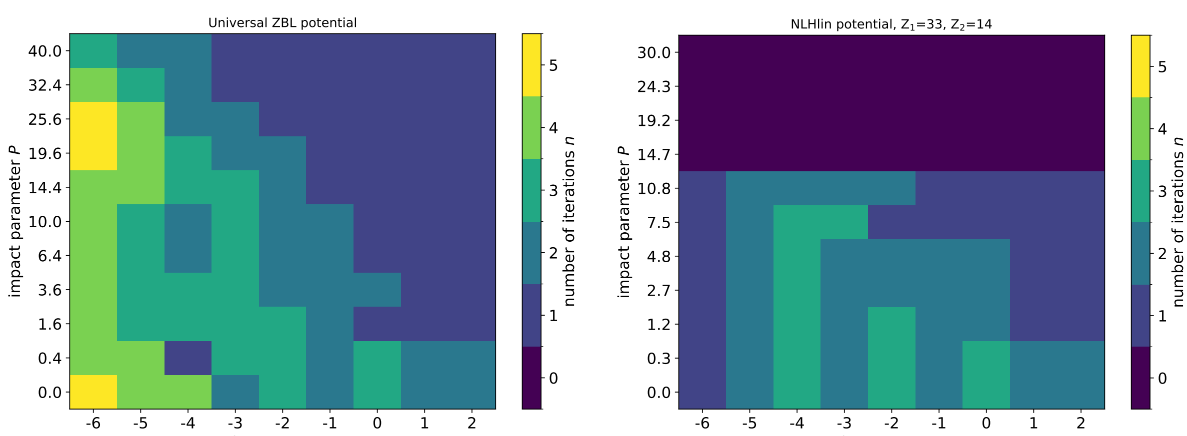

Figure 1.C.1: Number of iterations required to reach a relative accuracy of 10^{-3} in R_0 as a function of energy and impact parameter. Left: Universal ZBL potential. Right: NLHlin potential for As-Si using Eq. (1.16) as the screening length. The reduced impact parameters P have been chosen as to cover the same range of unscaled impact parameters p.

Figure 1.C.1 shows the number of iterations required to reach a relative accuracy of 10^{-3} between the last two calculated values of R_0 for the universal ZBL and the NLHlin

potential, using the approximations to $\Phi(R)$ described in this section for obtaining an initial condition. Usually this means that the last value is accurate to 10^{-6}, assuming quadratic convergence has been reached.

From Fig. 1.C.1 it is apparent that the NLHlin potential shows better convergence behavior than the ZBL potential at low projectile energies, while the convergence is similar for higher energies.

1.D Variants of Newtons Method for Obtaining the Apsis of Collision (legacy)

1.D.1 Iterating towards the Exact Solution

We consider three options for iteration:

Newton's method

Newton's method to solve Eq. (1.23) for $R = R_0$ is given by Eq. (1.29). The initial condition Eq. (1.28) guarantees convergence of the iterations for screening functions with $\Phi'(R) < 0$ and $\Phi''(r) > 0$.

Modified Newton's method

Newton's method applied to Eq. (1.23) can also be interpreted as constructing the tangent lines to the terms $\Phi(R)/\varepsilon$ and $R - P^2/R$ and intersecting the two tangents, see the dashed lines in Fig. 1.D.1.

Instead, one could also intersect the tangent to $\Phi(R)/\varepsilon$ with the function $R - P^2/R$. The resulting quadratic equation has the solution[8]

$$R_0^{(i+1)} = \frac{b + \sqrt{b^2 + 4ac}}{2a} \tag{1.D.1}$$

with

$$a = \varepsilon - \Phi'(R_0^{(i)}), \quad b = \Phi(R_0^{(i)}) - R_0^{(i)}\Phi'(R_0^{(i)}), \quad c = \varepsilon P^2 \tag{1.D.2}$$

As shown in Fig. 1.D.1, both methods warrant $R_0^{(i)} < R_0^{(i+1)} < R_0$ provided $R_0^{(i)} < R_0$, as long as $\Phi''(R) > 0$. In addition, Fig. 1.D.1 demonstrates that Eq. (1.D.1) is a better approximation to R_0 than Eq. (1.29). This comes at the expense of another square root evaluation.

[8]The other sign in front of the square root is not possible since all coefficients a, b, and c are positive, which would result in a negative value of $R_0^{(i+1)}$.

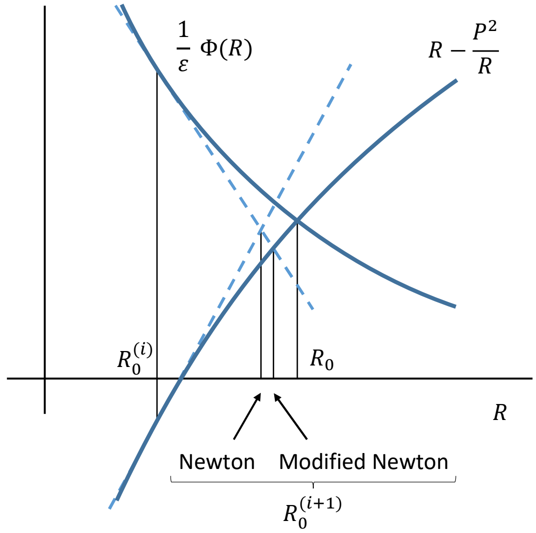

Figure 1.D.1: Illustration of Newton's method (linearization of both $\Phi(R)$ and $R - P^2/R$) and the method based on linearization of $\Phi(R)$ only. The latter comes closer to the actual solution R_0. For both methods $R_0^{(i)} < R_0^{(i+1)} < R_0$ provided $R_0^{(i)} < R_0$.

Newton's method applied to $\log(f(R))$

Instead of using Eq. (1.29) with $f(R)$ from Eq. (1.23), one can move the first and third term of $f(R)$ to the right-hand side of $f(R) = 0$, take the logarithm, and move the right-hand side back to the left-hand side. One then obtains

$$f_{\log}(R) = \log\left[\frac{1}{\varepsilon}\Phi(R) \Big/ \left(R - \frac{P^2}{R}\right)\right] = 0 \tag{1.D.3}$$

The function $f_{\log}(R)$ can be used with Newton's method Eq. (1.29). It maintains the properties $f'_{\log}(R) < 0$ and $f''_{\log}(R) > 0$ for screening functions that are given by a sum of exponentials. Therefore, Newton's method is guaranteed to converge if one starts with an initial condition to the left of the solution. The idea of taking the logarithm is to reduce the nonlinearity of $f(R)$, but obviously comes at the expense of the evaluation of another special function.

1.D.2 Convergence

Figure 1.D.2 evaluates the perfomance of the combinations of initial condition and iteration algorithm for the ZBL screening function. It shows the number of iterations required to reach a relative accuracy of 10^{-3} between the last two calculated values of R_0. Usually this means that the last value is accurate to 10^{-6}, assuming quadratic convergence has been reached.

In the left column, results for the initial condition Eq. (1.24) ($R_0^{(0)} = P$) are shown. As ex-

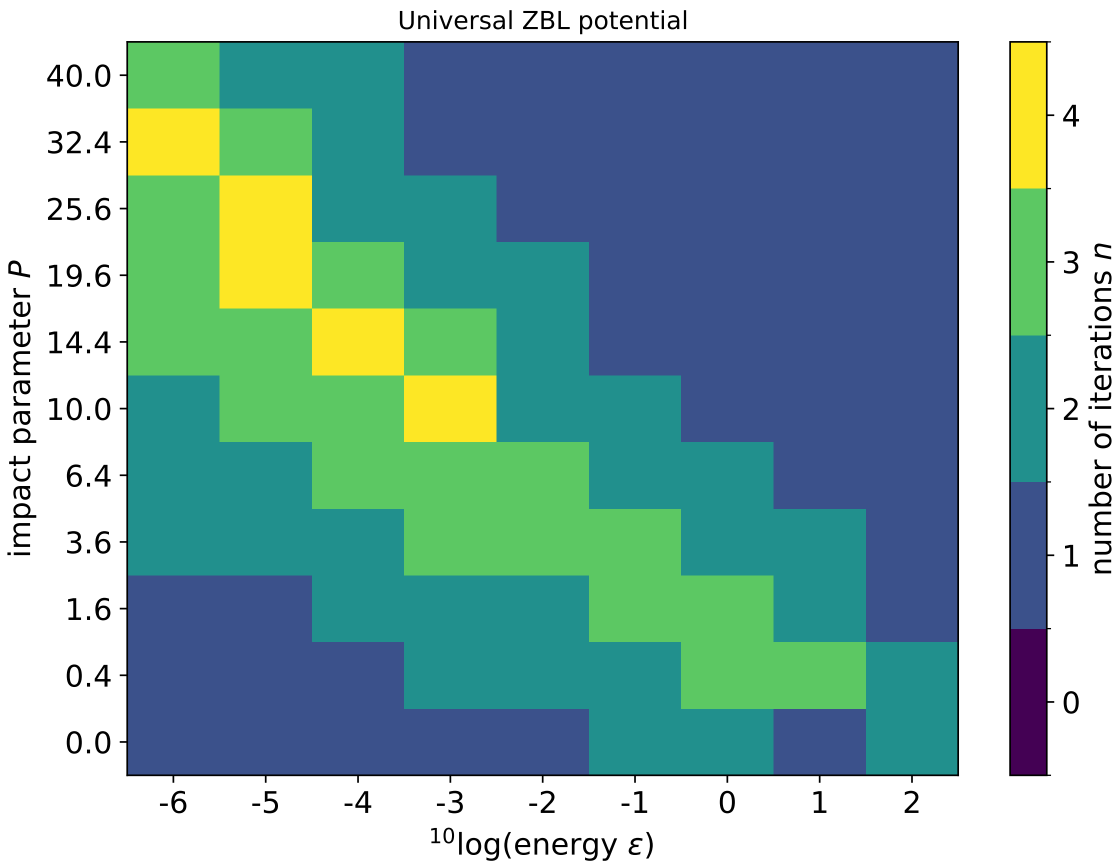

Figure 1.D.2: Number of iterations required to reach a relative accuracy of 10^{-3} in R_0 as a function of energy and impact parameter. Left column: using $R_0^{(0)} = P$ Eq. (1.24) as initial condition. Right column: using Eq. (1.28) as initial condition. The columns differ by the iteration method used: First row: Newton. Second row: Eq. (1.D.1) in the first iteration, then Newton. Third row: Eq. (1.D.1) for all iterations. Fourth row: Eq. (1.D.1) in the first iteration, then logarithmic Newton Eq. (1.D.3). The interatomic potential used was the universal ZBL potential.

pected, fewer iterations are required for large energies and large impact parameters (small deflection angles) than for small energies and small impact parameters (large deflection angles), since R_0 is closer to P. For large deflection angles, more than 10 iterations may be required using ordinary Newton or Eq. (1.D.1). Significant improvement is obtained by using the logarithmic function Eq. (1.D.3), limiting the number of iterations to 6. Note that Newton's method with the logarithmic function cannot be started with $R_0^{(0)} = P$, since this would lead to $f_{\log}(R \to P) \to \infty$. Therefore, Eq. (1.D.1) has been used in the first iteration.

In the right column, the table of $R_0^{(0,\varepsilon)}(\varepsilon)$ for head-on collisions has been used as described above. Compared to the left column, the number of iterations at low energies and small impact parameters is significantly reduced (not couting the iterations necessary to obtain the table). There are only slight differences in the iteration counts between the different iteration methods. Using Eq. (1.D.1) and Eq. (1.D.3) save one iteration in rare cases (compare first, second, and fourth row), while using Eq. (1.D.1) beyond the first collision has no benefit (compare second and third row). Since convergence is only marginally improved by using Eq. (1.D.1) and Eq. (1.D.3), while additional evaluations of a square root or a logarithm is necessary, using ordinary Newton's method seems to be sufficient.

In summary, when no table of $R_0^{(0,\varepsilon)}(\varepsilon)$ for head-on collisions is used, it is recommended to use Eq. (1.D.1) in the first iteration and Newton's method with the logarithmic function Eq. (1.D.3) for the remaining iterations. Significant acceleration is obtained by using the table of $R_0^{(0,\varepsilon)}(\varepsilon)$ for head-on collisions to obtain an initial condition. Then it is sufficient to use ordinary Newton's method. Note that the table needs to be calculated only once for a given interatomic potential.

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